Technical Note

Power Electronics and Control Systems

L(RC) Filter Design Guidelines:

grid-side filters for AFEs, machine-side filters for PSMs, and IMs

Topologies, sizing relations, resonance placement, damping and interaction with the current control

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Abstract: Design guidelines for the passive filters of two-level voltage-source converters: the L, LC, LCL and L(RC)L filter between an active front end (AFE) and a grid of given impedance, and the LC or L(RC) output filter between an inverter and an induction motor (IM) or a permanent-magnet synchronous motor (PMSM). The note first settles the switching ripple of the converter-side inductor, with exact closed forms for sinusoidal, space-vector and discontinuous PWM, the mean and the rms ripple over the fundamental period, and a comparison with the usual $V_{dc}/(8 f_{sw} L)$ rule. It then gives, for the grid side, the sizing of the converter inductor, of the capacitor (reactive power), of the grid inductor (attenuation at the switching frequency against IEEE 519), the placement of the resonance with respect to the grid frequency, the switching frequency and the critical frequency $f_s/6$ of the digital current loop, the passive damping options (series resistor, R–C branch) with their losses, and the interaction with the grid impedance. For the motor side it covers the reflected-wave overvoltage with long cables, the dv/dt and sine filter options, the resonance window, the loaded resonance set by the high-frequency inductance of the machine (transient inductance of the IM, L_d/L_q of the PMSM), the damping, the capacitor current and voltage budget, and the consequences for the current control and for sensorless (observer-based) control. Two worked examples, carried out step by step for one 450 kW, 480 V, 2 kHz drive (the L(RC)L grid filter of its active front end and the L(RC) sine filter of its induction or permanent-magnet motor), a comparison table AFE/IM/PMSM and a bibliography close the note; the numbers are reproduced by the companion Python script.

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Nomenclature

All quantities are per phase of the star equivalent of the three-phase system; RMS values unless a hat (peak) or a "pp" (peak-to-peak) is written. Per-unit values refer to the base quantities of the converter (Z_b, L_b, C_b). The same filter symbols are used on the grid side (Fig. 1a, inductances L_1, L_2) and on the motor side (Fig. 1b, inductance L_f); the damping-network symbols are common to both.

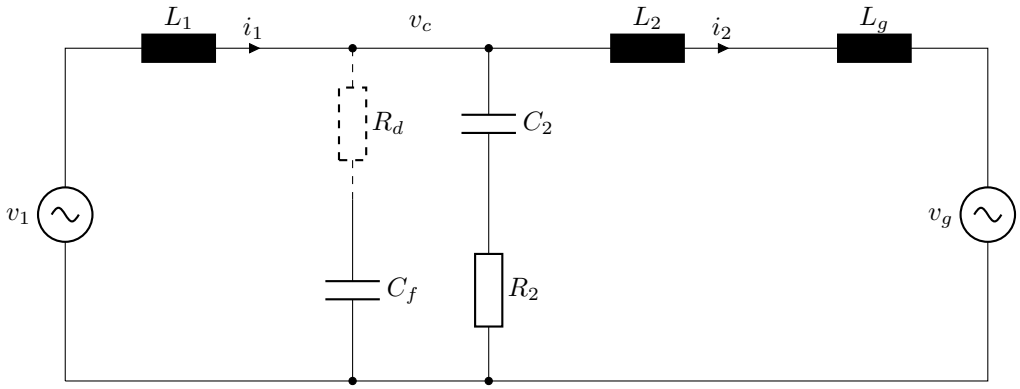
Symbol	Description	Unit
Converter, PWM and base quantities		
V_{dc}	DC-link voltage of the two-level voltage-source converter (VSC)	V
$f_{sw},\ \omega_{sw},\ T_{sw}$	switching (carrier) frequency, $\omega_{sw} = 2\pi f_{sw}$, switching period $1/f_{sw}$	Hz, rad/s, s
$f_s,\ T_s$	update (sampling) frequency of the digital current loop, $f_s = f_{sw}$ with single update and $2f_{sw}$ with double update; $T_s = 1/f_s$; the computation and PWM delay is $1.5T_s$	Hz, s
$f_1,\ \omega_1$	fundamental frequency (grid frequency on the grid side, motor supply frequency on the motor side), $\omega_1 = 2\pi f_1$	Hz, rad/s
$f_{1,max},\ \omega_{1,max}$	highest fundamental frequency of the drive (top speed, field weakening included)	Hz, rad/s
$V,\ V_{ph}$	RMS line-to-line voltage and phase (star) voltage, $V_{ph} = V/\sqrt{3}$	V
$S_n,\ P_n$	rated apparent power of the converter, rated active power of the motor	VA, W
$I_n,\ \hat{I}_n$	rated RMS phase current and its peak, $\hat{I}_n = \sqrt{2}I_n$	A
$\eta,\ \cos\varphi$	rated efficiency and power factor of the motor, $I_n = P_n/(\sqrt{3}V\eta\cos\varphi)$	–
$Z_b,\ L_b,\ C_b$	base impedance (V^2/S_n for the grid side, V_{ph}/I_n for the motor side), base inductance Z_b/ω_1 and base capacitance $1/(\omega_1 Z_b)$ at the rated fundamental frequency	Ω, H, F
m	modulation index: peak of the phase reference voltage over $V_{dc}/2$; linear range $m \le 1$ for SPWM, $m \le 2/\sqrt{3}$ for SVM/THI	–
θ	angle of the reference voltage vector; $\theta = 0$ when the phase-a voltage is at its peak, 90° at its zero crossing	rad
$v_{ref},\ v_{1,n}$	phase reference voltage and switched phase voltage with respect to the load neutral (0, $\pm V_{dc}/3$, $\pm 2V_{dc}/3$)	V
$T_1,\ T_2,\ T_0$	dwell times of the two active vectors and total dwell time of the zero vectors in one switching period; T_{0a}, T_{0b} are the parts of T_0 spent on 000 and on 111	s
SPWM, SVM, THI, DPWM	sinusoidal PWM (no zero sequence), space-vector PWM (equal split of the zero vectors), third-harmonic injection (same ripple as SVM), discontinuous PWM with 60° clamping (DPWM1)	–
k	ripple coefficient, $\Delta i = k\,V_{dc}/(f_{sw}L)$; tabulated in Table 2 as $1/x$	–
$\Delta i_{\mathrm{pp}},\ \Delta i_{\mathrm{pp},max}$	peak-to-peak switching ripple of the inductor current in one switching period, and its maximum over the fundamental period	A
$I_{rip,rms}$, HDF	RMS value of the switching ripple over the fundamental period and harmonic distortion factor of Holmes–Lipo, Eq. (2.5)	A, –
k_r	ripple allowance of the design, $\Delta i_{\mathrm{pp},max}/\hat{I}_n$	–
ϵ_L	relative fundamental voltage drop on the filter inductance(s), $\omega_1 L I_n/V_{ph}$	–
Converter losses and reactor model (Section 2.2)		
$E_{on},\ E_{off},\ E_{rr}$	switching energies of the IGBT (turn-on, turn-off) and of its diode (reverse recovery) at the datasheet reference I_{ref}, V_{ref}	J
I_{pk}	repetitive peak current, $\hat{I}_n + \Delta i_{\mathrm{pp},max}/2$	A
$A_p,\ A_c,\ A_w$	area product of the reactor and its factors, core section and window area	m^4, m^2

Symbol	Description	Unit
$B_{max},\ J,\ k_u$	peak flux density at I_{pk}, winding current density, window fill factor	T, A/m^2, –
$k_h,\ k_e$	hysteresis and eddy-current coefficients of the core loss per unit mass, $p = k_h f \hat{B}^2 + k_e f^2 \hat{B}^2$ (p in W/kg, f in Hz, $\hat{B}$ in T)	–
F_r	AC-to-DC resistance ratio of the winding at f_{sw}	–
$v_{L,hf}$	switching part of the voltage across the converter-side inductor; $\langle v_{L,hf}^2 \rangle$ its mean square	V
$m_L,\ P_L$	mass and total loss of the three-phase reactor (P_{cu}, $P_{fe,1}$, $P_{fe,sw}$: copper, core at f_1, core at f_{sw})	kg, W
$L_{opt},\ k_{r,opt}$	inductance and ripple allowance at the minimum of P_L	H, –
$c_P,\ c_m$	cost weights of the capitalised losses (EUR per W) and of the reactor mass (EUR per kg)	–
Grid-side filter (Fig. 1a)		
$L_1,\ L_2$	converter-side and grid-side filter inductances	H
$L_g,\ L_2'$	grid inductance at the point of common coupling (transformer and upstream network) and total grid-side inductance $L_2' = L_2 + L_g$	H
C_f	filter capacitance per phase, star equivalent (a delta bank uses $C_f/3$ per branch)	F
$r,\ x$	inductance ratio L_2/L_1 and capacitor fraction C_f/C_b (the capacitor reactive power is $Q_c = xS_n$)	–
$v_1,\ v_c,\ v_g$	converter, capacitor-node and grid phase voltages	V
$i_1,\ i_2$	converter-side and grid-side currents	A
$Z_1,\ Z_2,\ Z_c$	impedances of L_1, of L_2' and of the shunt network (capacitor with its damping)	Ω
$f_{res},\ \omega_{res}$	resonance frequency of the LCL filter, Eq. (3.2); f_{res}^{min}, f_{res}^{max} its values at the weakest and at the stiffest grid	Hz, rad/s
$f_2,\ \omega_2$	resonance of L_2' with C_f, $1/(2\pi\sqrt{L_2' C_f})$, the corner of the current ratio i_2/i_1	Hz, rad/s
k_a	attenuation of the filter at the switching frequency, $\lvert i_2/i_1 \rvert_{\omega_{sw}}$	–
$I_{1,sw},\ I_{2,sw}$	RMS converter-side and grid-side currents of the switching-frequency cluster	A
$I_1(f),\ I_2(f)$	RMS value of the spectral line at frequency f of i_1, i_2 (e.g. $f = f_{sw} \pm 2f_1$)	A
L_t	trap inductor in series with C_1, tuned at f_{sw} (LLCL filter)	H
$I_c,\ \hat{v}_{c,sw}$	RMS capacitor current and peak switching ripple of the capacitor voltage	A, V
S_{sc}, SCR	short-circuit power at the point of common coupling (PCC) and short-circuit ratio S_{sc}/S_n; I_{sc}/I_L is the ratio of IEEE 519 with I_L the maximum demand load current	VA, –
$u_k,\ S_T$	short-circuit (leakage) voltage and rating of the supply transformer	pu, VA
h, TDD	harmonic order and total demand distortion of IEEE 519	–
f_c	crossover frequency of the current loop	Hz
$f_s/6$	critical resonance frequency of the digital current loop (Parker–McGrath–Holmes)	Hz
$Y_c,\ Z_g$	converter output admittance and grid impedance of the impedance-based stability criterion	S, Ω
Damping networks (both sides)		
R_d	resistor in series with C_f (series damping)	Ω
$C_1,\ C_2,\ n$	main capacitor and damping-branch capacitor of the R–C network, $n = C_2/C_1$; $C_1 + C_2$ is the total capacitance	F, –

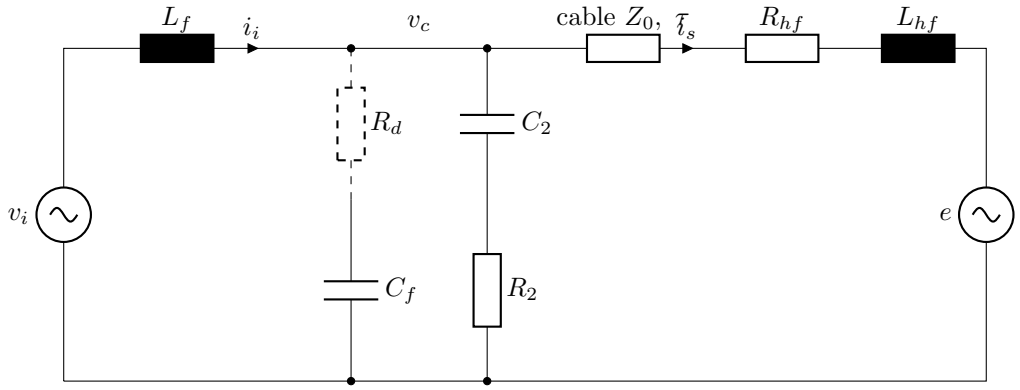
Symbol	Description	Unit
R_2	resistor of the R–C damping branch	Ω
L_p	inductance seen by the capacitor node: $L_1 \parallel L_2'$ on the grid side, $L_f \parallel L_{hf}$ on the motor side	H
$R_0,\ \omega_0$	characteristic impedance $\sqrt{L_p/C_1}$ and undamped resonance $1/\sqrt{L_pC_1}$ of the damped network (on the motor side $\omega_0 = 2\pi f_0$ with L_f)	Ω, rad/s
$\rho,\ \rho_Z,\ \rho_H,\ \rho_P$	normalised damping resistance R_2/R_0 and its closed-form optima (minimum peak impedance, minimum peak of the voltage transfer function, real pole at ω_0)	–
$\zeta,\ \zeta_m,\ Q$	damping ratio of the filter resonance, contribution of the machine to it, quality factor $1/(2\zeta)$	–
X_c	capacitor reactance at the switching frequency, $1/(\omega_{sw}C_f)$	Ω
$v_{c,sw}$	RMS switching-ripple voltage on the capacitor node	V
$P_{R_d},\ P_{R_2}$	losses of the damping resistors (fundamental plus switching terms)	W
Motor-side filter (Fig. 1b)		
$v_i,\ i_i$	inverter output voltage and current (filter input)	V, A
$v_c,\ v_s,\ i_s$	capacitor voltage, stator (motor terminal) voltage, $v_s = v_c$ without cable, and stator current	V, A
$L_f,\ R_{Lf}$	filter inductance and its series resistance	H, Ω
$L_{hf},\ R_{hf}$	inductance and resistance presented by the machine at the filter resonance: $L_{hf} = \sigma L_s = L_{ls} + L_{lr}'$ (transient inductance of the induction motor, stator and referred rotor leakage), $L_{hf} = L_d$ or L_q (PMSM); R_{hf} includes the deep-bar rotor resistance of the IM	H, Ω
e	back-EMF of the machine	V
$f_0,\ f_p$	unloaded resonance $1/(2\pi\sqrt{L_fC_f})$ and resonance with the machine connected, $f_0\sqrt{1 + L_f/L_{hf}}$	Hz
$Z_0,\ v,\ \tau,\ l$	surge impedance $\sqrt{L'/C'}$, propagation velocity, transit time l/v and length of the motor cable	Ω, m/μs, s, m
t_r	rise time (10–90 %) of the voltage edge at the cable input	s
$\Gamma_m,\ \Gamma_s$	voltage reflection coefficients at the motor and at the inverter end of the cable	–
l_{crit}	critical cable length $vt_r/2$ at which the full reflection is reached	m
$\hat{V}_m,\ OF$	peak voltage at the motor terminals and overshoot factor $\hat{V}_m/V_{dc} - 1$	V, –
$Z_m,\ C_{cable}$	high-frequency impedance of the motor seen by the cable, cable capacitance	Ω, F
G_m	equivalent conductance of the machine branch at f_p, $R_{hf}/(R_{hf}^2 + (2\pi f_p L_{hf})^2)$	S
I_c	fundamental capacitor current $\omega_1 C_f V_{ph}$	A
$\alpha_i,\ \alpha_v,\ \alpha_s$	bandwidths of the inverter-current, capacitor-voltage and stator-current loops of the cascaded control	rad/s
f_{inj}	frequency of the high-frequency injection of saliency-based sensorless control	Hz
MTPA, MTPV	maximum-torque-per-ampere and maximum-torque-per-volt trajectories of the PMSM	–

1 Scope

Two-level three-phase voltage-source converter (VSC) [16, 15] with DC link V_{dc}, switching frequency f_{sw}, double-edge regular-sampled PWM with single or double update. Two applications are treated with the same formulas: the grid side of an active front end (AFE), Fig. 1(a), and the output side of a motor inverter, Fig. 1(b). Formulas are quoted without derivation except where the derivation is short and not found in the references; the numbers of the worked examples are reproduced by the companion script `lrc_filter_example.py`.



(a) Grid side: L ($C_f = 0$), LCL, L(RC)L; L_g is the grid (transformer and network) inductance.



(b) Motor side: LC, L(RC); the machine is R_{hf}, L_{hf} (σL_s for the IM, L_d/L_q for the PMSM) and the back-EMF e.

Figure 1: Filter structures. The damping is either the series resistor R_d (dashed) or the parallel branch C_2–R_2 (the "RC" of L(RC)); only one of the two is used.

2 Switching ripple of the converter-side inductor

2.1 Ripple coefficients

The converter-side inductor (L_1 or L_f) is sized first from its switching ripple. With the capacitor node a stiff voltage at f_{sw} (LCL, LC) or a stiff grid (plain L), the phase current ripple over one switching period $T_{sw} = 1/f_{sw}$ is piecewise linear with slopes $(v_{1,n} - v_{ref})/L$, where $v_{1,n}$ is the converter phase voltage with respect to the load neutral (0, $\pm V_{dc}/3$, $\pm 2V_{dc}/3$) and v_{ref} the reference. Let $m = \hat{V}_{ref}/(V_{dc}/2)$ be the modulation index ($m \le 1$ for sinusoidal PWM, $m \le 2/\sqrt{3} = 1.155$ for space-vector or third-harmonic PWM). All ripple figures scale as

$$\Delta i = k \, \frac{V_{dc}}{f_{sw} L}, \tag{2.1}$$



and the coefficient k depends on the modulator, on m and on the position θ along the fundamental period. Two closed forms cover the two extremes (phase voltage at its peak, $\theta = 0$, and at its zero crossing, $\theta = 90°$):

$$k_{\mathrm{pp}}(\theta = 0) = \frac{m}{4}\Big(1 - \frac{m}{2}\Big) \text{ (SPWM)}, \qquad \frac{m}{4}\Big(1 - \frac{3m}{4}\Big) \text{ (SVM, THI)}, \qquad \frac{m}{2}\Big(1 - \frac{3m}{4}\Big) \text{ (DPWM, clamped phase)}, \tag{2.2}$$

$$k_{\mathrm{pp}}(\theta = 90°) = \frac{\sqrt{3}\,m}{12} = \frac{m}{6.93} \qquad \text{(all modulators)}. \tag{2.3}$$

Eq. (2.2) follows from $\Delta i_{\mathrm{pp}} = v_{ref}\,(T_0 + |T_{0a} - T_{0b}|)/(2L)$, with $T_0 = T_{0a} + T_{0b}$ the total zero-vector time and T_{0a}, T_{0b} its split between 000 and 111 (equal for SVM, $(1-m)T_{sw}/2$ and $(1-m/2)T_{sw}/2$ for SPWM at $\theta = 0$, all on one vector for the clamped phase of DPWM); at $\theta = 0$, $T_0/T_{sw} = 1 - 3m/4$. Eq. (2.3) follows from the two active vectors of equal dwell $T_1 = T_2 = (\sqrt{3}/4)\,m\,T_{sw}$ giving $\pm V_{dc}/3$ to the phase at its zero crossing. The peak-to-peak ripple is largest at the phase peak for small m and at the zero crossing for large m (crossover at $m = 0.56$ for SVM, 0.85 for SPWM). The worst case over the whole linear range is therefore

$$\Delta i_{\mathrm{pp},max} = \frac{V_{dc}}{4\sqrt{3}\,f_{sw}L} = \frac{V_{dc}}{6.93\,f_{sw}L} \qquad \text{(SPWM, at } m = 1\text{)}, \tag{2.4}$$

$$\Delta i_{\mathrm{pp},max} = \frac{V_{dc}}{6\,f_{sw}L} \qquad \text{(SVM/THI, at } m = 2/\sqrt{3}\text{; DPWM, at } m = 2/3 \text{ and } 2/\sqrt{3}\text{)}.$$

The RMS value of the ripple over the fundamental period, needed for the inductor and capacitor losses and for the harmonic current injected into the capacitor, is given by the harmonic distortion factor of Holmes and Lipo [15]:

$$I_{rip,rms} = \frac{V_{dc}}{24\,f_{sw}L}\sqrt{\mathrm{HDF}(m)}, \qquad \mathrm{HDF} = \begin{cases} \frac{3}{2}m^2 - \frac{4\sqrt{3}}{\pi}m^3 + \frac{9}{8}m^4 & \text{(SPWM)}, \\ \frac{3}{2}m^2 - \frac{4\sqrt{3}}{\pi}m^3 + \Big(\frac{27}{16} - \frac{81\sqrt{3}}{64\pi}\Big)m^4 & \text{(SVM)}. \end{cases} \tag{2.5}$$

Table 2 lists k for the maximum and the mean (over the fundamental period) peak-to-peak ripple and for the RMS ripple, computed exactly by the companion script; the closed forms (2.2)–(2.5) are reproduced to four digits.

Table 2: Ripple coefficient k of (2.1), written as $1/x$: maximum peak-to-peak over the fundamental period, mean peak-to-peak, RMS of the ripple. Two-level VSC, double-edge regular sampling, isolated neutral.

	SPWM			SVM / THI			DPWM1 (60° clamping)		
m	max pp	mean pp	RMS	max pp	mean pp	RMS	max pp	mean pp	RMS
0.4	1/12.5	1/17.6	1/67	1/14.3	1/18.7	1/68	1/7.1	1/10.4	1/35
0.6	1/9.5	1/13.7	1/52	1/11.6	1/15.3	1/55	1/6.1	1/8.8	1/30
0.8	1/8.3	1/12.1	1/44	1/8.7	1/14.4	1/49	1/6.3	1/8.9	1/31
0.9	1/7.7	1/11.4	1/41	1/7.7	1/14.0	1/47	1/6.8	1/9.4	1/32
1.0	1/6.9	1/10.9	1/37	1/6.9	1/13.5	1/45	1/6.9	1/10.4	1/35
1.1	–	–	–	1/6.3	1/13.0	1/42	1/6.3	1/11.5	1/37
1.155	–	–	–	1/6.0	1/12.8	1/40	1/6.0	1/11.9	1/38

Which ripple formula to use. The widely used $L = V_{dc}/(8 f_{sw} \Delta i_{\mathrm{pp}})$ is neither the maximum nor the mean: at $m \approx 1$ with SVM the true maximum is $V_{dc}/(6.9 f_{sw} L)$ (15 % above the rule), the worst case over m is $V_{dc}/(6 f_{sw} L)$ (33 % above, the value used by Liserre et al. [1] and Reznik et al. [2]), while the mean peak-to-peak over the fundamental is $V_{dc}/(13.5 f_{sw} L)$ (40 % below). The rule is a fair "typical period" estimate; it is not an upper bound. Recommended use: (i) size L and check the saturation current $\hat{I}_n + \Delta i_{\mathrm{pp},max}/2$ with



the maximum of Table 2 at the largest m of the application ($V_{dc}/(6f_{sw}L)$ when the converter reaches the hexagon, $V_{dc}/(7f_{sw}L)$ for an AFE at $m \approx 1$); (ii) use the RMS column, or (2.5), for the capacitor current, the damping losses and the harmonic current at f_{sw}; (iii) with discontinuous PWM at the same f_{sw} the ripple is up to twice that of SVM at low m, so compare at equal switching losses (f_{sw} raised by 1.5).

2.2 The ripple–inductance trade-off: where the optimum L lies

Eq. (2.1) ties L to the ripple allowance $k_r = \Delta i_{\mathrm{pp},max}/\hat{I}_n$ once f_{sw} is given, $L = k_{\mathrm{pp},max}V_{dc}/(f_{sw}k_r\hat{I}_n)$; which k_r to take is a balance between what the ripple costs in the converter and what the inductance costs in the reactor. The two sides are quantified in turn, then combined for the 450 kW drive at f_{sw} = 2–5 kHz (Fig. 2, Table 3); the same applies to the motor-side L_f, which sees the same switched voltage.

Converter side: f_{sw} is set by the semiconductor, L hardly matters. With the datasheet energies E_{on}, E_{off}, E_{rr} at I_{ref}, V_{ref} (linear in current and voltage), the switching loss of one IGBT–diode pair carrying a sinusoidal current is

$$P_{sw} = f_{sw}\,(E_{on} + E_{off} + E_{rr})\,\frac{\hat{I}_n}{\pi I_{ref}}\,\frac{V_{dc}}{V_{ref}}, \tag{2.6}$$

and P_{sw} plus the conduction loss at the overload current must fit the thermal budget of the module: this is what fixes f_{sw} (2–5 kHz with 1700 V IGBTs at this power). The ripple then enters the semiconductor in three ways. (i) *Peak current*: with continuous PWM the top IGBT turns on at the valley and off at the crest of the local ripple, so the repetitive peak is $\hat{I}_n + \Delta i_{\mathrm{pp},max}/2 = \hat{I}_n(1 + k_r/2)$ at rated current and $\hat{I}_n(1.5 + k_r/2)$ at 150 % overload; it must stay below the module I_{CRM}, below the over-current trip with its margin, and below the saturation current of L. This is the "$I_0 + \Delta I/2$" constraint, and it is the only one through which the IGBT sees L. (ii) *Switching loss*: E_{off} is paid at the crest and E_{on}, E_{rr} at the valley, so to first order

$$\frac{\Delta P_{sw}}{P_{sw}} = \frac{E_{off} - E_{on} - E_{rr}}{E_{on} + E_{off} + E_{rr}}\,\frac{\pi}{4}\,\frac{\Delta i_{\mathrm{pp},mean}}{\hat{I}_n} = 0.40\,k_r\,\frac{E_{off} - E_{on} - E_{rr}}{E_{on} + E_{off} + E_{rr}} \quad (\mathrm{SVM},\ m \approx 1), \tag{2.7}$$

with $k_{\mathrm{pp},mean}/k_{\mathrm{pp},max} = 0.51$; a few percent either way: for 1700 V IGBT4-class modules $E_{on} + E_{rr}$ exceeds E_{off} by 20–40 % of the total, so the ripple *lowers* P_{sw} by 3–6 % at $k_r = 0.4$; with E_{off}-dominated devices the sign reverses. (iii) *Conduction loss*: the RMS current rises by $(I_{rip,rms}/I_n)^2 = (0.22\,k_r)^2 < 1\,\%$. The semiconductor is therefore indifferent to L except through the peak; its stake in the choice is the trip margin, not the losses.

Reactor side: a scaling model. Consider a family of geometrically similar gapped reactors (laminated SiFe, foil or flat-wire winding with fill factor k_u, forced-air cooling) designed at the same peak flux density B_{max} at $I_{pk} = \hat{I}_n(1 + k_r/2)$ and at the same current density J. The area product

$$A_p = A_c A_w = \frac{L\,I_{pk}\,I_{rms}}{k_u B_{max} J}, \qquad I_{rms} = \sqrt{I_n^2 + I_{rip,rms}^2}, \tag{2.8}$$

fixes the linear dimension $\propto A_p^{1/4}$, so core and copper masses scale as $A_p^{3/4} \propto L^{3/4}$ (the factor $(1 + k_r/2)^{3/4}$ from I_{pk} is weak). The losses are:

- copper: $P_{cu} = R_{dc}\,(I_n^2 + F_r I_{rip,rms}^2) = \rho_{cu} J^2 V_{cu}\,(1 + F_r I_{rip,rms}^2/I_n^2) \propto L^{3/4}$, with $F_r = R_{ac}/R_{dc}$ at f_{sw} (3–5 for a foil winding; the ripple term is a few percent);
- core at the fundamental: $P_{fe,1} = m_{fe}\,(k_h f_1 + k_e f_1^2)\,B_1^2 \propto L^{3/4}$, with $B_1 = B_{max}/(1 + k_r/2)$ and the loss separation $p = k_h f \hat{B}^2 + k_e f^2 \hat{B}^2$ per unit mass ($k_e = \pi^2\sigma d^2/(6\rho_m)$ for the lamination thickness d, valid while d stays below the skin depth of the steel);
- core at the switching frequency: the eddy-current term depends on $\langle (dB/dt)^2 \rangle = \langle v_{L,hf}^2 \rangle/(NA_c)^2$ with

$NA_c = LI_{pk}/B_{max}$, the hysteresis term on the amplitude of the ripple flux, so

$$P_{fe,sw} = m_{fe}\left[\frac{k_e}{2\pi^2}\,\langle v_{L,hf}^2\rangle\left(\frac{B_{max}}{L\,I_{pk}}\right)^2 + k_h f_{sw}\hat{B}_{sw}^2\right], \qquad \hat{B}_{sw} = \sqrt{3}\,B_{max}\,\frac{I_{rip,rms}}{I_{pk}}. \tag{2.9}$$

Here $\langle v_{L,hf}^2\rangle$ is the mean square of the switching part of the voltage across the inductor (converter phase voltage minus reference, the capacitor node being stiff at f_{sw}). For the two-level VSC with SVM it has the closed form

$$\frac{\langle v_{L,hf}^2\rangle}{V_{dc}^2} = \frac{1}{4} - \frac{\langle v_0^2\rangle}{V_{dc}^2} - \frac{m^2}{8} = \frac{2\sqrt{3}}{3\pi}\,m - \frac{m^2}{8} = 0.057 \quad (m = 1.02:\ \text{RMS } 0.24\,V_{dc}), \tag{2.10}$$

where $\langle v_0^2\rangle = V_{dc}^2\,[a/36 + (1-a)/4]$ is the mean square of the common-mode voltage and $a = 3\sqrt{3}\,m/(2\pi)$ the fraction of the period spent on the active vectors (the script reproduces it from the exact waveform). Two facts follow. The switched voltage does not depend on f_{sw}, so at a given L the eddy-current loss, the part of $P_{fe,sw}$ that grows when L is reduced, is the same at 2 kHz and at 5 kHz and scales as $m_{fe}/L^2 \propto L^{-5/4}$. And it is 2.4 times what a triangular ripple with the same RMS at f_{sw} would produce, because the multi-slope SVM waveform has a far larger $\langle v^2\rangle$ than its current ripple suggests: a core loss estimated "from the ripple at f_{sw}" underestimates it by that factor.

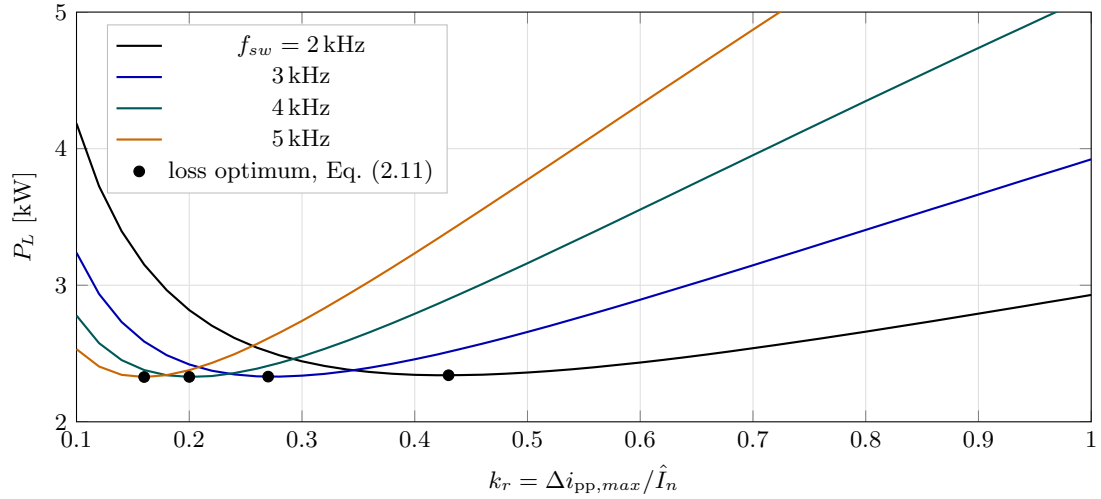
The optimum. The total reactor loss $P_L = P_{cu} + P_{fe,1} + P_{fe,sw}$ is of the form $AL^{3/4} + BL^{-5/4}$, with its minimum where

$$P_{fe,sw} = \frac{3}{5}\,(P_{cu} + P_{fe,1}), \qquad L_{opt} = L_t\sqrt{\frac{r_t}{0.6}}, \quad r_t = \left.\frac{P_{fe,sw}}{P_{cu} + P_{fe,1}}\right|_{L_t}, \tag{2.11}$$

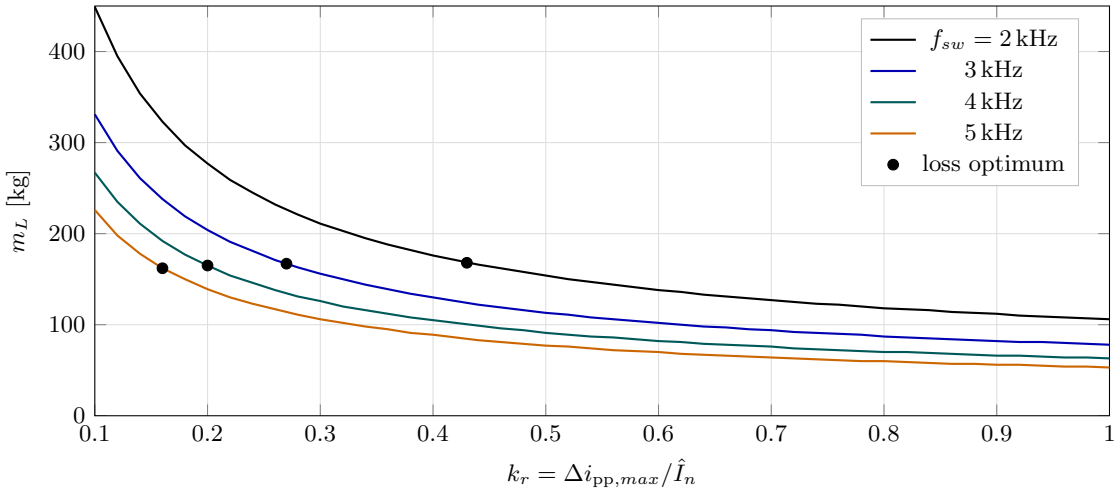
i.e. where the switching-frequency core loss is 60 % of the rest. The second form gives L_{opt} from the loss split of any trial design L_t, e.g. from a supplier's loss sheet, provided the switching-frequency core loss is specified for the PWM voltage (2.10) and not for a sinusoidal ripple. With a Steinmetz law $p \propto f^{\alpha}\hat{B}^{\beta}$ in place of the eddy term the ratio is $(3/4)/(\beta - 3/4)$ and $L_{opt} = L_t\,[r_t(4\beta-3)/3]^{1/\beta}$; with the thermally consistent scaling $J \propto A_p^{-1/8}$ (larger reactors run at a lower current density) the ratio becomes 1/2 and L_{opt} moves up by about 10 %. The mass $m_L \propto [(1+k_r/2)/k_r]^{3/4}$ falls monotonically with k_r, so every design with $k_r < k_{r,opt}$ is dominated (heavier *and* more lossy) and the useful range is $k_r \geq k_{r,opt}$: the loss optimum when the losses are capitalised (with $c_P \approx$ 5–10 EUR/W over the life of the drive against $c_m \approx$ 10–15 EUR/kg of reactor, minimising $c_P P_L + c_m m_L$ moves k_r up by only 10 %), a larger k_r when the cabinet volume is binding, up to the peak-current limit of the converter and the capacitor-current and attenuation limits of Sections 4 and 5. Since $\langle v_{L,hf}^2\rangle$ does not depend on f_{sw}, neither does L_{opt}: raising f_{sw} at the optimum lowers the ripple ($k_{r,opt} \propto 1/f_{sw}$) but leaves mass and loss of the reactor unchanged; the smaller reactor expected from a higher f_{sw} is obtained only by keeping k_r, i.e. by moving off the optimum and paying core loss, or with thinner laminations and better steel ($k_e \propto d^2$, hence $k_{r,opt} \propto 1/d$), amorphous or powder cores.

Numbers for the 450 kW drive. AFE of Example 1 (I_n = 601 A, V_{dc} = 800 V, SVM at m = 1.02: $k_{\mathrm{pp},max}$ = 1/6.8, k_{rms} = 1/44); reactor model: 0.35 mm non-oriented steel (3 W/kg at 1.5 T, 50 Hz: k_h = 0.024 W/(kg Hz T^2), $k_e = 5.3\times10^{-5}$ W/(kg Hz^2 T^2)), B_{max} = 1.4 T, J = 3 A/mm^2, k_u = 0.3, UI core with a square leg a and window $0.8a \times 2.5a$, three single-phase reactors, $F_r = 2 + f_{sw}/2\,\text{kHz}$. Fig. 2 gives P_L and m_L against k_r for f_{sw} = 2, 3, 4, 5 kHz, Table 3 the optima. At 2 kHz the loss optimum is k_r = 0.43 (L = 161 µH, 13 %), a flat minimum (0.37–0.50 within 1 %) on which the k_r = 0.40 of step A3 of Example 1 sits; the conventional 0.25 would cost 10 % more loss and 42 % more mass. At 4 kHz the optimum is k_r = 0.20 with the same L = 173 µH: 0.25, the allowance of the 690 V design of Example 1b, costs 2 % of loss and saves 14 % of mass; 0.40 saves 36 % of mass for 20 % more loss. The reactor at the optimum is the same at the four

frequencies, $L \approx 170\,\mu\text{H}$ (14 %), 165 kg, 2.3 kW (0.47 % of S_n), with $P_{fe,sw}/(P_{cu} + P_{fe,1}) = 0.6$; the higher f_{sw} pays on the filter side, where the grid line scales as $1/f_{sw}^3$ and the capacitor shrinks. The peak current at the optimum is $\hat{I}_n(1 + k_r/2) = 1033\,\text{A}$ at 2 kHz and 936 A at 4 kHz (1460 A and 1361 A at 150 % overload), within the repetitive peak rating of the 1400 A-class 1700 V modules used at this power. Rule for the 2–5 kHz range with 0.35 mm steel: $k_{r,opt} \approx 0.8\,\text{kHz}/f_{sw}$, equivalently $L_{opt} \approx 13$–$14\,\%$ of L_b, to be moved up with better steel and checked with (2.11) on the supplier's loss split.



(a) Total loss P_L of the three-phase reactor.



(b) Mass m_L of the three-phase reactor (core and copper).

Figure 2: Ripple–inductance trade-off of the converter-side reactor of the 450 kW AFE ($I_n = 601\,\text{A}$, $V_{dc} = 800\,\text{V}$, SVM) against the ripple allowance k_r, for $f_{sw} = 2, 3, 4, 5\,\text{kHz}$. The dots mark the loss optimum (2.11), which lies at the same $L \approx 170\,\mu\text{H}$ for the four frequencies. Reactor model of Section 2.2: 0.35 mm SiFe, $B_{max} = 1.4\,\text{T}$, $J = 3\,\text{A/mm}^2$, $k_u = 0.3$; mass decreasing monotonically, loss with a minimum.

Table 3: Loss optimum of the converter-side reactor of the 450 kW AFE (model of Fig. 2); P_L and m_L for the three phases; the last column is the loss and mass penalty of the conventional $k_r = 0.25$ with respect to the optimum.

f_{sw}	$k_{r,opt}$	L_{opt}	I_{pk}	$P_{cu}+P_{fe,1}$	$P_{fe,sw}$	P_L	m_L	k_r within	k_r at the	$k_r = 0.25$:
kHz	–	µH (% L_b)	A	W	W	W (% S_n)	kg	1 % of $P_{L,min}$	cost optimum	ΔP_L, Δm_L
2	0.43	161 (13.2)	1033	1436	905	2341 (0.47)	168	0.37–0.50	0.47	+10 %, +42 %
3	0.27	171 (14.0)	965	1450	880	2331 (0.47)	167	0.24–0.32	0.30	+0.3 %, +5 %
4	0.20	173 (14.2)	936	1444	884	2329 (0.47)	165	0.18–0.23	0.22	+2 %, −14 %
5	0.16	173 (14.2)	919	1434	894	2328 (0.47)	162	0.14–0.18	0.17	+9 %, −26 %

3 Filter structures and transfer functions

With $Z_1 = sL_1$, $Z_2 = s(L_2 + L_g)$ and Z_c the impedance of the shunt network of Fig. 1(a), and the grid voltage a short at the switching harmonics,

$$\frac{i_1}{v_1} = \frac{Z_c + Z_2}{Z_1 Z_2 + Z_c(Z_1 + Z_2)}, \qquad \frac{i_2}{v_1} = \frac{Z_c}{Z_1 Z_2 + Z_c(Z_1 + Z_2)}, \qquad \frac{i_2}{i_1} = \frac{Z_c}{Z_c + Z_2}. \tag{3.1}$$

For the undamped LCL ($Z_c = 1/(sC_f)$), writing $L_2' = L_2 + L_g$,

$$\frac{i_2}{v_1} = \frac{1}{s^3 L_1 L_2' C_f + s(L_1 + L_2')}, \qquad \omega_{res}^2 = \frac{L_1 + L_2'}{L_1 L_2' C_f}, \qquad \frac{i_2}{i_1} = \frac{1}{1 + s^2 L_2' C_f}, \qquad \omega_2^2 = \frac{1}{L_2' C_f}. \tag{3.2}$$

At the switching frequency the attenuation of the LCL with respect to its own converter current is

$$\begin{aligned} k_a &= \left|\frac{i_2}{i_1}\right|_{\omega_{sw}} = \frac{1}{\omega_{sw}^2 L_2' C_f - 1} \approx \left(\frac{f_2}{f_{sw}}\right)^2, \qquad \text{equivalently [1]} \\ \frac{i_2(\omega_{sw})}{i_L(\omega_{sw})} &= \frac{1}{\left|1 + r\,(1 - \omega_{sw}^2 L_1 C_f)\right|}, \quad r = \frac{L_2}{L_1}, \end{aligned} \tag{3.3}$$

the second form being referred to the current $i_L = v_1/(j\omega_{sw} L_1)$ of the plain L filter; the two coincide where C_f is a short at f_{sw}. The damped networks are $Z_c = R_d + 1/(sC_f)$ (series resistor) and $Z_c = \frac{1}{sC_1} \parallel \left(R_2 + \frac{1}{sC_2}\right)$ (R–C branch, with $C_1 + C_2$ the total capacitance). The series resistor turns the −60 dB/decade roll-off of i_2/v_1 into −40 dB/decade above $1/(2\pi R_d C_f)$ and degrades (3.3) to $k_a = \sqrt{R_d^2 + X_c^2}/\sqrt{R_d^2 + (\omega_{sw} L_2' - X_c)^2}$, $X_c = 1/(\omega_{sw} C_f)$; the R–C branch keeps the roll-off and is analysed in [14, 11, 12, 13]. The motor-side LC filter of Fig. 1(b) obeys the same algebra with $Z_1 = sL_f$ and $Z_2 = R_{hf} + sL_{hf}$ (the back-EMF is a short at the ripple frequencies, the cable adds its capacitance to C_f):

$$\frac{v_c}{v_i} = \frac{Z_c \parallel Z_2}{Z_1 + Z_c \parallel Z_2}, \qquad f_0 = \frac{1}{2\pi\sqrt{L_f C_f}}, \qquad f_p = f_0\sqrt{1 + \frac{L_f}{L_{hf}}}, \qquad \left|\frac{v_c}{v_i}\right|_{f_{sw}} \approx \frac{1}{(f_{sw}/f_p)^2 - 1}, \tag{3.4}$$

where f_0 is the unloaded resonance and f_p the resonance with the machine connected.



4 Grid-side filter of an active front end

4.1 Design inputs

Rated power S_n, voltage V, frequency f_1, DC-link voltage V_{dc} (boost operation requires $V_{dc} \geq \sqrt{2}V(1+\epsilon_L)$, i.e. $m \leq 2\sqrt{2}V_{ph}(1+\epsilon_L)/V_{dc}$ with ϵ_L the filter drop), f_{sw} and f_s, the current-feedback choice (converter current i_1 or grid current i_2), the range of the grid impedance at the point of common coupling (PCC), and the harmonic limits. The grid inductance per phase, transformer included, is

$$L_g = \frac{V^2}{\omega_1 S_{sc}} = \frac{u_k V^2}{\omega_1 S_T} + L_g^{MV}, \qquad \frac{S_{sc}}{S_n} = \frac{I_{sc}}{I_L} \text{ (short-circuit ratio)}, \tag{4.1}$$

to be bracketed between the stiffest ($S_{sc,max}$) and the weakest ($S_{sc,min}$) expected PCC. The limits of IEEE 519 [17] at the PCC depend on I_{sc}/I_L (Table 4); the design practice for a stand-alone converter is to apply them to its own I_n. At a high pulse ratio the switching cluster lies above $h = 50$ and the last column is used for the whole cluster; at a low pulse ratio ($f_{sw}/f_1 < 50$) the cluster straddles the rows and each of its lines, $f_{sw} \pm 2f_1$ first, is checked against the limit of its own order (Section 6.1). IEC 61800-3 [18] sets the EMC environment (C1–C4) and refers to the IEC 61000-3 series for the low-frequency emission.

Table 4: IEEE 519-2014 current distortion limits at the PCC, 120 V–69 kV, in percent of the maximum demand current I_L (odd harmonics; even harmonics 25 % of these).

I_{sc}/I_L	$h < 11$	$11 \leq h < 17$	$17 \leq h < 23$	$23 \leq h < 35$	$35 \leq h \leq 50$	TDD
< 20	4.0	2.0	1.5	0.6	0.3	5.0
20–50	7.0	3.5	2.5	1.0	0.5	8.0
50–100	10.0	4.5	4.0	1.5	0.7	12.0
100–1000	12.0	5.5	5.0	2.0	1.0	15.0
> 1000	15.0	7.0	6.0	2.5	1.4	20.0

4.2 Converter-side inductor L_1

From Section 2, with the ripple allowance $k_r = \Delta i_{\mathrm{pp},max}/\hat{I}_n$ (15–30 % is usual at 4–5 kHz and 40–50 % at 2 kHz, the loss optimum of the reactor scaling as $1/f_{sw}$, Section 2.2; a smaller value increases L_1, its mass, its copper loss and the voltage drop, a larger one the peak current, the core loss and the capacitor current),

$$L_1 \geq \frac{k\, V_{dc}}{f_{sw}\, k_r \hat{I}_n}, \qquad k = \frac{1}{7} \text{ (SVM, } m \approx 1), \quad \frac{1}{6} \text{ (worst case)}, \tag{4.2}$$

and the drop $\epsilon_L = \omega_1(L_1 + L_2)I_n/V_{ph}$ (typically 0.05–0.15 pu in total) must be compatible with V_{dc}. The inductor carries I_n plus the ripple, must not saturate at $\hat{I}_n + \Delta i_{\mathrm{pp},max}/2$ (plus the overload and the inrush at pre-charge), and its core loss is set by the ripple RMS of Table 2.

4.3 Capacitor C_f

The capacitor is bounded by its reactive power at the fundamental, which the converter must compensate (and which raises the no-load current),

$$C_f \leq x\, C_b = \frac{x\, S_n}{\omega_1 V^2}, \qquad Q_c = \omega_1 C_f V^2 = x S_n, \qquad x \leq 0.05 \text{ [1]}, \tag{4.3}$$



(C_f is the star value; a delta bank uses $C_f/3$ per branch). A larger C_f lowers L_2 for a given attenuation and lowers f_{res}; with x = 2–5 % the capacitor is the cheapest element and is normally taken at its upper bound. The capacitor must withstand $1.1V$ plus the switching ripple voltage $\hat{v}_{c,sw} \approx I_{rip,rms}\sqrt{2}/(\omega_{sw}C_f)$, carry $I_c = \sqrt{(\omega_1 C_f V_{ph})^2 + I_{rip,rms}^2}$ and be rated for the harmonic background of the grid (5th, 7th) that it absorbs through L_2'.

4.4 Grid-side inductor L_2 and attenuation

The converter-side current at the switching frequency is estimated from the ripple RMS of Table 2: the first carrier cluster holds 85–94 % of it with SVM, split mainly between the two lines at $f_{sw} \pm 2f_1$, each about $V_{dc}/(90 f_{sw} L_1)$ at $m \approx 1$ and $V_{dc}/(73 f_{sw} L_1)$ at full modulation (Table 5; the analytical sideband spectrum is in [15]). With the admissible grid harmonic $I_{2,sw}$ of Table 4 at the order of the line considered, the required attenuation is $k_a = I_{2,sw}/I_{1,sw}$ and, from (3.3) with $L_g = 0$ (worst case: L_g only adds attenuation),

$$L_2 \geq \frac{1 + 1/k_a}{\omega_{sw}^2 C_f}, \qquad r = \frac{L_2}{L_1} = 0.2\text{–}1. \tag{4.4}$$

Since $I_{1,sw} \propto 1/L_1$ and $k_a \approx 1/(\omega_{sw}^2 L_2' C_f)$, the grid harmonic scales as $1/(f_{sw}^3 L_1 L_2' C_f)$ and the total inductance for a given harmonic is minimum at $L_2' = L_1 + 1/(\omega_{sw}^2 C_f) \approx L_1$; values of r above unity give little further attenuation for the volume [1, 3]. The grid inductor may be the leakage of the dedicated transformer or a line reactor. If the attenuation cannot be met, raise C_f up to its bound (it helps only as $1/\sqrt{C_f}$ at fixed total inductance), then L_1, then f_{sw}; at low pulse ratios ($f_{sw}/f_1 \lesssim 40$) a trap inductor $L_t = 1/(\omega_{sw}^2 C_f)$ in series with the capacitor (LLCL filter [23]) notches the first cluster and is the usual way out, see Example 1.

4.5 Resonance placement

With L_g between L_g^{min} and L_g^{max} the resonance (3.2) sweeps the band $[f_{res}^{min}, f_{res}^{max}]$, f_{res}^{min} at the weakest grid. The constraints are

$$10 f_1 \leq f_{res}^{min}, \qquad f_{res}^{max} \leq \frac{f_{sw}}{2}\ [1], \qquad f_{res} \notin \left[0.8, 1.2\right]\frac{f_s}{6}\ [6]. \tag{4.5}$$

The lower bound keeps the resonance away from the low-order grid harmonics and from the current-loop crossover f_c (design $f_c \leq f_{res}/3$, with $f_c \approx f_{sw}/20 \ldots f_{sw}/10$ for a PI in the dq frame [7]); the upper bound preserves the attenuation at f_{sw}. The third constraint comes from the discrete-time analysis of the current loop with the computation and PWM delay ($1.5T_s$): with converter-current feedback the LCL resonance is stable without any damping only for $f_{res} > f_s/6$, with grid-current feedback only for $f_{res} < f_s/6$, and at $f_{res} = f_s/6$ no controller gain stabilises the loop [6]. Place the whole band $[f_{res}^{min}, f_{res}^{max}]$ on the side consistent with the feedback variable, or provide damping (passive, Section 4.6, or active) whenever the band crosses or approaches $f_s/6$ or when the grid impedance is uncertain [8, 5].

4.6 Passive damping

Series resistor. The classical choice [1] is one third of the capacitor reactance at the resonance,

$$R_d = \frac{1}{3\omega_{res} C_f}, \qquad \zeta \approx \frac{R_d}{2}\sqrt{\frac{C_f}{L_p}} \approx \frac{1}{6}, \qquad L_p = \frac{L_1 L_2'}{L_1 + L_2'}, \qquad P_{R_d} = 3R_d\left[(\omega_1 C_f V_{ph})^2 + I_{rip,rms}^2\right], \tag{4.6}$$

where the loss has a fundamental term (the whole capacitor current flows in R_d) and a switching term; both are usually 0.1–0.3 % of S_n [4]. The price is the attenuation penalty of Section 3.



R–C branch. Adding to the main capacitor C_1 a branch $C_2 = nC_1$ with R_2 in series keeps the −60 dB/decade slope; C_1 is the capacitor that sets f_{res} and the attenuation, and the branch is added to it rather than carved out of a fixed total, because splitting raises the shunt impedance at f_{sw} by $(1+n)$ and costs attenuation whenever f_{res} is within an octave of f_{sw} (Example 1). With $R_0 = \sqrt{L_p/C_1}$ and $\rho = R_2/R_0$, the three closed-form choices of [14] are

$$\begin{aligned} \rho_Z &= \sqrt{\frac{(2+n)(4+3n)}{2n^2(4+n)}} && \text{minimum peak of the output impedance [11],} \\ \rho_H &= \frac{\sqrt{(1+n)(2+n)/2}}{n} && \text{minimum peak of } |v_c/v_1|, \\ \rho_P &= \frac{2+n}{2n} && \text{real pole at } \omega_0 = 1/\sqrt{L_pC_1}, \end{aligned} \tag{4.7}$$

with $\zeta = n/(2\sqrt{2(2+n)})$ for ρ_P (0.20 for $n=1$, 0.35 for $n=2$) and a peak of $|v_c/v_1|$ not lower than $(2+n)/n$. On the AC grid the branch carries the fundamental capacitive current of C_2, so

$$P_{R_2} = 3R_2\frac{(\omega_1 C_2 V_{ph})^2}{1+(\omega_1 C_2 R_2)^2} + 3\frac{v_{c,sw}^2}{R_2}, \qquad v_{c,sw} \approx \frac{I_{rip,rms}}{\omega_{sw}C_1}, \tag{4.8}$$

which limits n to 1–2 and makes the loss comparable with the series resistor [4]. Both networks are to be checked at f_{res}^{min} and f_{res}^{max}; in modern AFEs passive damping is kept small as a robustness backstop and the bulk of the damping is active (capacitor-current or capacitor-voltage feedback [7, 5, 10]).

4.7 Interaction with the grid impedance and stability

The grid sees the converter as a current source with an output admittance $Y_c(s)$ shaped by the current controller; stability of the interconnection is that of the loop $Z_g(s)Y_c(s)$ [9]. With the digital delay the real part of Y_c becomes negative between about $f_s/6$ and $f_s/2$ for converter-current control, so any resonance of Z_g with the filter falling in that band is destabilised, whatever the gains; passive damping and the choice of f_{res} of Section 4.5 move the resonance out of the band or make Y_c passive there. Practical rules: (i) evaluate f_{res} and the phase margin at both ends of the L_g range; (ii) a weak grid (SCR < 3) also demands a slow PLL (≤ 20 Hz) and a reduced current-loop bandwidth; (iii) the background distortion of the grid is amplified by the LCL between f_2 and f_{res}, which is why $f_{res} \geq 10 f_1$ and why the capacitor must be rated for the 5th/7th harmonics; (iv) verify the design with the transformer leakage and the cable of the actual installation, not with $L_g = 0$ only.

5 Output filters between inverter and motor

5.1 Why: dv/dt, reflected waves and insulation stress

A PWM edge with rise time t_r travels along the motor cable (surge impedance $Z_0 = \sqrt{L'/C'} \approx 20$–$80\,\Omega$, velocity $v \approx 150$–200 m/µs, transit time $\tau = l/v$) and is reflected at the motor with coefficient $\Gamma_m = (Z_m - Z_0)/(Z_m + Z_0)$, close to +1 because the high-frequency impedance of the winding is $\gg Z_0$ (0.8–0.95 for motors below 100 kW), and at the inverter with $\Gamma_s \approx -1$. The terminal voltage reaches $V_{dc}(1+\Gamma_m)$ when the edge is shorter than the round trip [24, 25]:

$$l_{crit} = \frac{v\,t_r}{2}, \qquad \hat{V}_m \leq V_{dc}\Big[1 + \Gamma_m \min\Big(1, \frac{l}{l_{crit}}\Big)\Big], \tag{5.1}$$

the second expression being the design envelope of [26] (the lossless-line solution for a linear edge lies below it; real edges and cable losses smooth it). Double pulsing and polarity reversal within the cable ringing period



4τ raise the peak up to $3V_{dc}$ [27]; the high-frequency models needed for prediction are in [28]. With IGBT edges of 0.1–0.3 µs, l_{crit} is 10–30 m and any industrial cable gives the full $2V_{dc}$. The limits are those of the motor insulation: NEMA MG 1 Part 31 [22] (definite-purpose inverter-fed motors) allows $3.1\,V_{rated}$ peak with $t_r \geq 0.1$ µs for ≤ 600 V motors, Part 30 (general purpose) 1000 V peak and $t_r \geq 2$ µs; IEC 60034-18-41 [20] qualifies the insulation by stress category (overshoot factor and impulse rise time), IEC TS 60034-25 [21] gives the application guide and IEC TS 61800-8 [19] the method to specify the voltage at the power interface. The common-mode dv/dt (bearing currents, EMI) is a separate issue, mitigated by common-mode chokes and not covered here.

5.2 Filter options

- **Output reactor** (1–3 %): slows the edge seen by the cable only through the cable capacitance, $t_r \approx 2.2\sqrt{LC_{cable}}$-like; mostly limits the capacitive charging current and the inverter stress; insufficient alone above 50–100 m.
- dv/dt **filter** (LC, $f_0 = 5$–$10\,f_{sw}$, damped): limits the edge at the filter output to $t_r \approx 0.3/f_0$ (undamped 10–90 % rise; 0.3–0.5 with damping). From (5.1), for an admissible overshoot $OF = \hat{V}_m/V_{dc} - 1$ and cable length l: $l_{crit} \geq l\,\Gamma_m/OF$, i.e. $t_r \geq 2l\Gamma_m/(v\,OF)$. The capacitor is charged and discharged twice per switching period, so irrespective of the damping resistor the loss is bounded by $P \approx 3f_{sw}C_f V_{dc}^2$: C_f must stay at 0.1–0.5 µF and the filter cannot be made slow. Damping $\zeta = 0.3$–0.5 with R_d in series with C_f or a resistor across L_f (carries no fundamental); clamping diodes to the DC bus recover the energy [29, 26].
- **Sine filter** (LC or L(RC), $f_0 \ll f_{sw}$): sinusoidal motor voltage, no reflection problem at any length (the edge is $\sim 1/(2f_0)$), lower motor noise and harmonic losses, standard motors usable, but fundamental drop, capacitor current, reduced control bandwidth and field-weakening range [30, 29, 33]. The remainder of this section is on the sine filter.
- **Motor-terminal RC** ($R = Z_0$, $C \approx 3t_r/Z_0$): matches the cable at the motor end [25]; used when the filter cannot be at the inverter.

5.3 Sine filter: resonance window

The unloaded resonance f_0 must sit between the highest fundamental and its low-order harmonics on one side, and the switching frequency on the other:

$$(8\ldots10)\,f_{1,max} \leq f_0 \leq \frac{f_{sw}}{4\ldots3}, \qquad \left|\frac{v_c}{v_i}\right|_{f_{sw}} \approx \left(\frac{f_p}{f_{sw}}\right)^2 = -20\text{–}-28\,\text{dB}. \tag{5.2}$$

The lower bound keeps the 5th and 7th harmonics generated by the dead time and by the inverter nonlinearity (at $5f_1$, $7f_1$, up to 1.4 kHz for a 200 Hz machine) away from the resonance, where they would be amplified by $Q = 1/(2\zeta)$, and keeps the current-loop crossover below $f_0/3$. The window exists only if $f_{sw} \geq (24\ldots40)\,f_{1,max}$: a 50 Hz induction motor in field weakening up to 75 Hz is served by $f_{sw} = 4$ kHz, a high-speed or high-pole PMSM at 200 Hz needs $f_{sw} \geq 6$–8 kHz or a relaxed attenuation. This, with the capacitor current, is the main reason why sine filters are heavier for PMSM drives.



5.4 Sine filter: inductor, capacitor, loaded resonance

Inductor. Sized by the fundamental drop at $f_{1,max}$ and checked for the inverter ripple (the allowance follows Section 2.2 as on the grid side: $k_{r,opt} \propto 1/f_{sw}$, 0.4–0.5 at 2 kHz),

$$L_f \le \frac{\epsilon_L V_{ph}}{\omega_{1,max} I_n}, \quad \epsilon_L = 0.05\text{–}0.10; \qquad \Delta i_{\mathrm{pp},max} = \frac{V_{dc}}{6 f_{sw} L_f} \quad \text{(the inverter reaches } m = 2/\sqrt{3} \text{ at } f_{1,max}\text{)}, \tag{5.3}$$

With $L_f = 3\text{–}6\,\%$ the inverter current ripple is 25–45 % of $\hat{I}_n$: it flows entirely into C_f, the motor sees none of it, but the inverter must be rated for the peak and for the switching losses, and the inductor core for the ripple RMS of Table 2. The drop reduces the voltage available to the machine by $\omega_1 L_f I_n$ (vectorially): it must be budgeted in V_{dc} or accepted as an earlier field-weakening onset.

Capacitor. From the chosen f_0 and the fundamental capacitive current,

$$C_f = \frac{1}{(2\pi f_0)^2 L_f}, \qquad I_c = \omega_{1,max} C_f V_{ph} \le (0.1 \ldots 0.2)\, I_n, \qquad \boldsymbol{i}_i = \boldsymbol{i}_s + j\omega_1 C_f \boldsymbol{v}_c, \quad \boldsymbol{v}_i = \boldsymbol{v}_c + (R_{Lf} + j\omega_1 L_f)\, \boldsymbol{i}_i, \tag{5.4}$$

the last two relations giving the inverter current and voltage that set the inverter rating and shift the torque-maximising trajectories of the machine [33]. The capacitor is rated for $1.1V$ plus the switching ripple $\hat{v}_{c,sw} \approx \sqrt{2} I_{rip,rms}/(\omega_{sw} C_f)$ and for $\sqrt{I_c^2 + I_{rip,rms}^2}$.

Loaded resonance. With the machine connected the resonance moves from f_0 to $f_p = f_0\sqrt{1 + L_f/L_{hf}}$ of (3.4), where L_{hf} is the inductance the machine presents at kilohertz frequencies: the transient (total leakage) inductance $\sigma L_s = L_{ls} + L'_{lr}$ of the induction motor, 0.15–0.25 pu, or L_d, L_q of the PMSM, 0.1–0.3 pu for surface magnets and 0.3–0.8 pu for interior magnets with $L_q > L_d$ and a strong dependence on i_q (saturation). The shift is 4–20 % and must be included in (5.2); for an IPM it differs between the d and q axes, so the plant seen by the current controller is anisotropic. The damping contributed by the machine is small because $L_{hf} \gg L_f$ decouples it from the capacitor node: the machine branch acts at f_p as a conductance $G_m = R_{hf}/(R_{hf}^2 + (2\pi f_p L_{hf})^2)$ and gives $\zeta_m = \frac{1}{2} G_m \sqrt{L_p/C_f}$ with $L_p = L_f \parallel L_{hf}$, i.e. a few 10^{-3} even with the deep-bar rotor resistance of the IM (0.05–0.15 pu at 1 kHz), and 10^{-4} for the PMSM, whose rotor has no cage. The resonance therefore has $Q \approx 10\text{–}20$, set by the inductor losses, unless it is damped explicitly.

5.5 Sine filter: damping

The same two passive networks of Section 4.6 apply with $\omega_{res} \to 2\pi f_p$ and $L_p = L_f \parallel L_{hf}$; on the motor side the fundamental reaches $f_{1,max}$ and the capacitor is a larger fraction of C_b, so the fundamental term dominates the loss of both networks and the R–C branch, which adds $C_2 = nC_f$ to the capacitance, costs typically twice the series resistor. The series resistor $R_d = 1/(3 \cdot 2\pi f_p C_f)$ ($\zeta \approx 0.17$) is the usual passive choice at 0.5–1 % of P_n; it also reduces the attenuation by the factor $\sqrt{1 + (2\pi f_{sw} R_d C_f)^2}$. Active damping (a virtual resistor through capacitor-voltage or capacitor-current feedback, or the cascaded control of the next section) is lossless and is preferred where the control has the necessary measurements; the passive resistor is then kept small as a backstop for the V/f mode and for the resonance excited by the dead-time harmonics.

5.6 Consequences for the control, IM versus PMSM

With the filter the inverter current i_i and voltage v_i are no longer the stator quantities: $i_s = i_i - C_f\, dv_c/dt$ and $v_s = v_c = v_i - (R_{Lf} + L_f\, d/dt)\, i_i$. A vector controller that ignores the filter commits a steady-state error of 5–15 % on the voltage (the drop on L_f) and of 5–20 % on the current (the capacitor current), and has the undamped LC resonance inside its loop. The structures that work are:



- **Cascaded control** [31]: inner loop on i_i (plant L_f, bandwidth $\alpha_i \leq 2\pi f_{sw}/10$), middle loop on v_c (plant C_f, $\alpha_v \approx \alpha_i/3 \ldots \alpha_i/5$), outer loop on i_s (plant L_{hf}, $\alpha_s \approx \alpha_v/3 \ldots \alpha_v/5$); the inner loops damp the resonance actively. The stator-current bandwidth ends at $f_p/5$–$f_p/3$, i.e. 100–400 Hz instead of the 0.5–1 kHz of a filterless drive, and the speed loop is correspondingly slower; this is the price for servo-like PMSM applications.
- **Full-order observer** of the filter and machine states from i_i (and v_i reference), which restores i_s, v_s and the flux without extra sensors [31, 32]; the observer must include L_f, C_f and the correct L_{hf} (for the IPM, L_d and L_q with their saturation).
- **Direct measurement** of i_s or v_c after the filter when the cabinet layout allows it.

Sensorless control. Back-EMF (fundamental-model) observers of the PMSM need v_s: using v_i instead of v_c injects the L_f drop directly into the position estimate, so the filter model must be in the observer [32]; the same holds for the IM flux observer [31]. Saliency-based high-frequency injection (f_{inj} = 0.5–2 kHz) is attenuated and phase-shifted by the filter and, near f_p, the response is dominated by the filter rather than by the saliency: keep $f_{inj} \leq f_p/3$ with the filter transfer function compensated, or measure i_s directly; many drives disable injection with a sine filter.

IM versus PMSM, summary. (i) L_{hf}: σL_s for the IM, L_d/L_q for the PMSM, larger and anisotropic for the IPM, so the loaded resonance shift and the plant differ per axis. (ii) The IM rotor cage adds a lossy branch but, as shown, the damping it gives to the filter is marginal; the PMSM gives none. (iii) The PMSM usually has the higher $f_{1,max}$, which narrows the window (5.2) and raises the required f_{sw}. (iv) The filter drop and the capacitor current change the voltage and current limits: for the PMSM they shift the MTPA/MTPV trajectories and anticipate the field weakening [33]; for the IM they reduce the maximum torque in field weakening. (v) The benefit differs too: for the PMSM the sinusoidal current removes the magnet eddy-current losses and the demagnetisation risk from the switching ripple at high temperature; for the IM it removes the rotor-bar harmonic losses and the acoustic noise.

6 Worked examples: 450 kW drive, 480 V, 2 kHz

The two examples size the AC sine filters of one drive: a 450 kW motor at 480 V, 60 Hz, fed by a two-level inverter through an L(RC) sine filter, with the DC link supplied by a 500 kVA AFE through an L(RC)L filter from a 480 V, 60 Hz grid. Both converters switch at f_{sw} = 2 kHz with SVM and double update (f_s = 4 kHz, delay $1.5T_s$ = 375 µs) and share V_{dc} = 800 V. The pulse ratio $f_{sw}/f_1 = 33.3$ is low: this is the case where the filter, not the converter, dominates the design, and the examples show where the general relations of Sections 4 and 5 bite. Each step gives the relation, the substitution and the result; the companion script reproduces every number.

6.1 Common data: ripple spectrum at pulse ratio 33

At a low pulse ratio the carrier clusters are close to the fundamental ($f_{sw} \pm 2f_1$ is $h = 31.3$ and 35.3), so the harmonic limits must be applied line by line rather than to the cluster as a whole. The exact ripple waveform of Section 2 over three fundamental periods (100 switching periods) gives the spectrum of Table 5: with SVM the first cluster carries 85–94 % of the ripple RMS, its two largest lines are at $f_{sw} \pm 2f_1$ (the carrier itself and $f_{sw} \pm 3kf_1$ are common mode and cancel in the line current), and the second cluster is 2–3.5 times smaller.

Table 5: Ripple RMS coefficients k of (2.1), written as $1/x$, by carrier cluster and by spectral line, SVM, $f_{sw}/f_1 = 33.3$ (computed by FFT of the exact piecewise-linear ripple).

	total	1st cluster	2nd cluster	$f_{sw} \pm 2f_1$		$f_{sw} \pm 4f_1$		$2f_{sw} \pm f_1$	$2f_{sw} \pm 5f_1$	
m	RMS	(f_{sw})	($2f_{sw}$)	−	+	−	+	±	−	+
1.00 (AFE)	1/45.0	1/53.1	1/98.7	1/90.2	1/94.1	1/127	1/137	1/155	1/328	1/336
1.02 (AFE with drop)	1/44.4	1/51.1	1/104	1/86.8	1/90.7	1/121	1/131	1/175	1/312	1/322
1.155 (motor, full V)	1/39.8	1/42.2	1/144	1/71.2	1/75.4	1/99.0	1/110	1/391–1/405	1/252	1/277

6.2 Example 1: grid-side L(RC)L sine filter of the AFE

Inputs. $V = 480\,\text{V}$, $f_1 = 60\,\text{Hz}$, $S_n = 500\,\text{kVA}$ (the 450 kW motor at $\eta = 0.96$ plus margin), $V_{dc} = 800\,\text{V}$, $f_{sw} = 2\,\text{kHz}$, $f_s = 4\,\text{kHz}$, SVM, converter-current (i_1) feedback, unity power factor at the grid, $S_{sc} = 12$–$48\,\text{MVA}$ at the PCC.

A1 – Base quantities and grid inductance. From (4.1):

$$I_n = \frac{S_n}{\sqrt{3}V} = 601\,\text{A}, \qquad \hat{I}_n = \sqrt{2}I_n = 851\,\text{A}, \qquad Z_b = \frac{V^2}{S_n} = 0.461\,\Omega,$$
$$L_b = \frac{Z_b}{\omega_1} = 1.222\,\text{mH}, \qquad C_b = \frac{1}{\omega_1 Z_b} = 5.76\,\text{mF}, \qquad \frac{I_{sc}}{I_L} = \frac{S_{sc}}{S_n} = 24\ldots96,$$
$$L_g = \frac{V^2}{\omega_1 S_{sc}} = \frac{480^2}{377 \cdot (12\ldots48) \cdot 10^6} = 50.9\text{–}12.7\,\mu\text{H}\ (4.2\text{–}1.0\,\%).$$

At the weakest PCC ($I_{sc}/I_L = 24$) the IEEE 519 row $20 < I_{sc}/I_L < 50$ applies: 1.0 % of I_n for $23 \le h < 35$ (6.0 A) and 0.5 % for $35 \le h \le 50$ (3.0 A).

A2 – Modulation index and ripple coefficients. The drop on $L_1 + L_2$ (30 %, step A5) is in quadrature with the current at unity power factor, so

$$m = \frac{2\sqrt{2}\,V_{ph}\sqrt{1+\epsilon_L^2}}{V_{dc}} = \frac{2\sqrt{2} \cdot 277 \cdot 1.04}{800} = 1.02,$$
$$k_{\text{pp},max} = \frac{\sqrt{3}\,m}{12} = 0.148 = \frac{1}{6.8}\ (2.3), \qquad k_{\text{pp},mean} = \frac{1}{13.4}, \qquad k_{rms} = \frac{1}{44.4}\ (2.5),$$

and from Table 5 the lines at $f_{sw} \pm 2f_1$ have $k = 1/86.8$ and $1/90.7$, those at $2f_{sw} \pm f_1$ $k = 1/175$.

A3 – Converter-side inductor L_1. From (4.2) with the usual allowance $k_r = 0.25$,

$$L_1 \ge \frac{k_{\text{pp},max} V_{dc}}{f_{sw} k_r \hat{I}_n} = \frac{0.148 \cdot 800}{2000 \cdot 0.25 \cdot 851} = 278\,\mu\text{H} = 22.7\,\%,$$

which is too large at 2 kHz: since the capacitor takes the whole ripple, the allowance is relaxed to $k_r = 0.40$ (the usual compromise of large converters at low f_{sw}, and the loss optimum of the reactor at 2 kHz, Section 2.2), giving $L_1 \ge 174\,\mu\text{H}$ and

$$L_1 = 180\,\mu\text{H}\ (14.7\,\%): \qquad \Delta i_{\text{pp},max} = \frac{0.148 \cdot 800}{2000 \cdot 180 \cdot 10^{-6}} = 328\,\text{A}\ (39\,\%\ \hat{I}_n), \qquad \Delta i_{\text{pp},mean} = 166\,\text{A},$$
$$I_{rip,rms} = \frac{800}{44.4 \cdot 2000 \cdot 180 \cdot 10^{-6}} = 50\,\text{A}\ (8.3\,\%\ I_n),$$
$$I_1(f_{sw}+2f_1) = \frac{800}{90.7 \cdot 2000 \cdot 180 \cdot 10^{-6}} = 24.5\,\text{A}, \qquad I_1(f_{sw}-2f_1) = \frac{800}{86.8 \cdot 2000 \cdot 180 \cdot 10^{-6}} = 25.6\,\text{A},$$

with the saturation current $\hat{I}_n + \Delta i_{\text{pp},max}/2 = 1015\,\text{A}$. For comparison, the worst case $V_{dc}/(6 f_{sw} L_1) = 370\,\text{A}$ and the rule $V_{dc}/(8 f_{sw} L_1) = 278\,\text{A}$.

A4 – Capacitor C_1. From (4.3) with $x = 0.05$:

$$C_1 \le 0.05\,C_b = 288\,\mu\text{F} \;\Rightarrow\; C_1 = 280\,\mu\text{F star (93}\,\mu\text{F delta)},$$
$$Q_c = \omega_1 C_1 V^2 = 24.3\,\text{kvar}\ (4.9\,\%\ S_n), \qquad I_c = \omega_1 C_1 V_{ph} = 29.3\,\text{A}.$$

A5 – Grid-side inductor L_2 from the attenuation. The binding line is $f_{sw} + 2f_1 = 2120\,\text{Hz}$ ($h = 35.3$, limit 3.0 A); with $L_g = 0$ in (4.4) (worst case) and then with the actual L_g:

$$k_a \le \frac{3.0}{24.5} = 0.123\ (-18.2\,\text{dB}), \qquad L_2' \ge \frac{1 + 1/k_a}{\omega^2 C_1} = \frac{9.13}{(2\pi \cdot 2120)^2 \cdot 280 \cdot 10^{-6}} = 184\,\mu\text{H};$$
$$f_{sw} - 2f_1 = 1880\,\text{Hz}: \qquad k_a \le \frac{6.0}{25.6} = 0.235, \qquad L_2' \ge 135\,\mu\text{H}.$$

Because $I_1 \propto 1/L_1$ and $k_a \approx 1/(\omega^2 L_2' C_1)$, the total inductance for a given I_2 is minimum at $L_2' = L_1 + 1/(\omega^2 C_1) \approx L_1$ ($r \approx 1$); here $L_1 = 172\,\mu\text{H}$, $L_2' = 192\,\mu\text{H}$, i.e. 30 % in total. The chosen $L_2 = 180\,\mu\text{H}$ ($r = 1$) gives

$$L_g^{min}: \quad L_2' = 192.7\,\mu\text{H}, \qquad k_a = \frac{1}{\omega^2 L_2' C_1 - 1} = 0.117,$$
$$I_2(f_{sw} + 2f_1) = 0.117 \cdot 24.5 = 2.86\,\text{A}\ (0.47\,\%), \qquad I_2(f_{sw} - 2f_1) = 3.92\,\text{A}\ (0.65\,\%);$$
$$L_g^{max}: \quad L_2' = 230.9\,\mu\text{H}, \qquad k_a = 0.095, \qquad I_2(f_{sw} + 2f_1) = 2.34\,\text{A}\ (0.39\,\%),$$

both lines within their limits. The drop of $L_1 + L_2 = 360\,\mu\text{H} = 29.5\,\%$ is in quadrature with the current:

$$V_1 = \sqrt{V_{ph}^2 + \big(\omega_1 (L_1 + L_2) I_n\big)^2} = \sqrt{277^2 + 81.6^2} = 289\,\text{V}\ (+4.2\,\%), \qquad m = \frac{2\sqrt{2} \cdot 289}{800} = 1.02,$$

well inside the SVM range at $V_{dc} = 800\,\text{V}$.

Why the filter is heavy at 2 kHz. The grid line $I_2(f_{sw} \pm 2f_1)$ scales as $1/(f_{sw}^3 L_1 L_2' C_1)$: at pulse ratio 33 the 0.5 % line limit at $h = 35$ with the capacitor at 5 % needs $L_1 L_2' \approx (15\,\%)^2$, and a larger capacitor helps only as $1/\sqrt{C_1}$. The three ways out, in order of practice: (i) a trap inductor $L_t = 1/(\omega_{sw}^2 C_1) = 22.6\,\mu\text{H}$ in series with C_1 (LLCL filter [23]), which notches the first cluster; the second cluster ($2f_{sw} \pm f_1$, 12.7 A at $L_1 = 180\,\mu\text{H}$) then sizes L_2 through the inductive divider $|Z_c|/|Z_c + Z_2|$ with $Z_c = j0.437\,\Omega$ at 4060 Hz, giving $L_2' \ge 55\,\mu\text{H}$; (ii) $f_{sw} = 4\,\text{kHz}$, which at the same k_r and C_1 gives $L_1 = 87\,\mu\text{H}$ and $L_2' \ge 51\,\mu\text{H}$, 11 % in total instead of 30 %; (iii) the limit applied at the plant PCC with the actual maximum demand current I_L of IEEE 519 rather than with I_n of the converter. A fourth option, the L(RC) without L_2 and with a large capacitor, needs a high pulse ratio and is reviewed in Section 6.3. The example continues with the plain L(RC)L to show the resonance and damping steps.

A6 – Resonance. From (3.2) and (4.5):

$$f_{res} = \frac{1}{2\pi}\sqrt{\frac{L_1 + L_2'}{L_1 L_2' C_1}} = \frac{1}{2\pi}\sqrt{\frac{372.7 \cdot 10^{-6}}{180 \cdot 192.7 \cdot 280 \cdot 10^{-18}}} = 986\,\text{Hz}\ (L_g^{min}),$$
$$f_{res} = 946\,\text{Hz}\ (L_g^{max}), \qquad 1003\,\text{Hz}\ (L_g = 0);$$
$$10 f_1 = 600\,\text{Hz} \le f_{res} \le \frac{f_{sw}}{2} = 1000\,\text{Hz}; \qquad \frac{f_s}{6} = 667\,\text{Hz}.$$

The band 946–986 Hz is above the critical $f_s/6$ (forbidden band 533–800 Hz): with i_1 feedback the loop is inherently stable [6] and the passive damping is a backstop against the uncertainty of L_g; the corner $f_2 =$

$1/(2\pi\sqrt{L_2'C_1}) = 626\,\text{Hz}$ at L_g^{max} is also above $10f_1$.

A7 – Damping, R–C branch added to C_1. With $L_p = L_1 \parallel L_2' = 93\,\mu\text{H}$ (L_g^{min}) and $R_0 = \sqrt{L_p/C_1} = 0.577\,\Omega$, the pole-placement choice ρ_P of (4.7) and the loss (4.8) give, for the three usual values of $n = C_2/C_1$:

n	C_2	total C	ρ_P	R_2	ζ	P_{R_2} fund. + sw.	P_{R_2}/S_n	k_a at $f_{sw}+2f_1$	$I_2(f_{sw}+2f_1)$
0.25	70 µF	6.1 %	4.50	2.59 Ω	0.059	414 + 136 = 550 W	0.11 %	0.112	2.73 A
0.5	140 µF	7.3 %	2.50	1.44 Ω	0.112	920 + 235 = 1155 W	0.23 %	0.108	2.64 A
1	280 µF	9.7 %	1.50	0.86 Ω	0.204	2202 + 370 = 2572 W	0.51 %	0.102	2.50 A

series resistor instead of the branch: $R_d = 1/(3\omega_{res}C_1) = 192\,\text{m}\Omega$, $\zeta = 0.17$, $P_{R_d} = 3R_d(I_c^2 + I_{rip,rms}^2) = 1940\,\text{W}$ (0.39 %); attenuation penalty $k_a = 0.143$, $I_2(f_{sw}+2f_1) = 3.50\,\text{A} > 3.0\,\text{A}$: rejected

The branch is added to C_1 rather than carved out of it: the attenuation and f_{res} stay those of C_1 (the branch even adds shunt admittance at f_{sw}, hence the slightly better k_a), whereas splitting a fixed total into C_1 and C_2 would raise the shunt impedance at f_{sw} by $(1+n)$ and, with f_{res} only one octave below f_{sw}, lose the limit. The series resistor loses it for the same reason. Selected: $n = 0.25$, $C_2 = 70\,\mu\text{F}$, $R_2 = 2.59\,\Omega$, $\zeta = 0.06$ ($Q \approx 8$), 550 W; the total capacitance is 6.1 % of C_b (30 kvar), the price of the backstop. $n = 0.5$ is the alternative where more passive damping is wanted against a weak or resonant grid.

A8 – Ratings and control. L_1: 601 A RMS, saturation $\geq 1015\,\text{A}$, ripple 50 A RMS at 2 kHz (core loss). L_2: 601 A line reactor, or the leakage of a dedicated transformer. C_1: $\geq 1.1V = 528\,\text{V}$ AC, switching ripple $\hat{v}_{c,sw} = 17\,\text{V}$, $\sqrt{I_c^2 + I_{1,sw}^2} = 52\,\text{A}$ RMS. C_2: 7.3 A; R_2: 2.59 Ω, 550 W for the three phases. Control: $f_c \leq f_{res}^{min}/3 = 315\,\text{Hz}$ ($f_{sw}/6.3$), phase margin to be verified at both L_g with the 375 µs delay; PLL $\leq 20\,\text{Hz}$ at $S_{sc,min}$.

6.3 Example 1b: L(RC) without L_2, the grid as second inductance

The lighter and very common alternative to Example 1 is the L(RC) filter alone, with the transformer and network inductance L_g as the second inductance of the current divider and a capacitor larger than the 5 % bound. It is reviewed here on a field design for a 690 V, 250 kW, 4 kHz AFE ($S_n = 275\,\text{kVA}$, $I_n = 230\,\text{A}$, $V_{dc} = 1100\,\text{V}$, $f_1 = 50\,\text{Hz}$, double update $f_s = 8\,\text{kHz}$, SVM, i_1 feedback): $L_1 = 500\,\mu\text{H}$, $C = 200\,\mu\text{F}$ with $R_d = 50\,\text{m}\Omega$ in series, $L_g \approx 50\,\mu\text{H}$ ($S_{sc} = 30\,\text{MVA}$, $I_{sc}/I_L = 110$). The criterion that such a filter must meet, with the *stiffest* grid expected (L_g^{min}, where the divider is weakest and the resonance highest), is

$$f_2 = \frac{1}{2\pi\sqrt{L_g C}}, \qquad f_{res} = f_2\sqrt{1+\frac{L_g}{L_1}} \approx f_2, \qquad 1.2\,\frac{f_s}{6} \leq f_{res}(L_g^{max}) \quad \text{and} \quad f_{res}(L_g^{min}) \leq \frac{f_{sw}}{2.5} \quad (k_a \leq 0.19), \tag{6.1}$$

with the line limit then checked through (3.3). Since L_g is only 0.5–2 %, (6.1) can be met only with a large capacitor and a high pulse ratio ($f_{sw}/f_1 \gtrsim 60$): it is the "large capacitor" design, whose resonance is wherever the grid puts it.

D1 – Base quantities and ripple. $Z_b = 1.73\,\Omega$, $L_b = 5.51\,\text{mH}$, $C_b = 1.84\,\text{mF}$; $m = 1.045$, $k_{\text{pp},max} = 1/6.6$, $k_{rms} = 1/43.7$; at pulse ratio 80 the lines $f_{sw} \pm 2f_1$ have $k = 1/84.5$ and 1/86.2. With (4.2) and $k_r = 0.25$, $L_1 \geq 1100/(6.6 \cdot 4000 \cdot 0.25 \cdot 325) = 510\,\mu\text{H}$: the 500 µH (9.1 %) are exactly the 25 % allowance (on the flat part of the loss curve of Section 2.2, whose optimum at 4 kHz is $k_r \approx 0.2$),

$$\Delta i_{\text{pp},max} = \frac{1100}{6.6 \cdot 4000 \cdot 500 \cdot 10^{-6}} = 83\,\text{A}\ (25\,\%), \qquad I_{rip,rms} = \frac{1100}{43.7 \cdot 4000 \cdot 500 \cdot 10^{-6}} = 12.6\,\text{A},$$
$$I_1(f_{sw} \pm 2f_1) = 6.5\text{–}6.4\,\text{A}, \qquad \hat{I}_n + \frac{\Delta i_{\text{pp},max}}{2} = 367\,\text{A}.$$

D2 – Capacitor. $C = 200\,\mu\text{F} = 10.9\,\%\,C_b$, twice the bound of (4.3): $Q_c = 30\,\text{kvar}$ (11 % of S_n, the no-load

current of the converter), $I_c = 25\,\text{A}$, $v_{c,sw} = 2.2\,\text{V}$. The size is not a choice but the consequence of D3.

D3 – Attenuation with the grid only. From (3.3) with the series resistor and the limits 1.15 A (0.5 %) and 2.3 A (1.0 %, the row of this grid):

L_g	f_2	f_{res}	f_{res}/f_{sw}	$k_a(f_{sw}+2f_1)$	$I_2(f_{sw}+2f_1)$	ζ from R_d
25 µH (0.5 %)	2251 Hz	2306 Hz	0.58	0.44	2.83 A (1.23 %)	0.072
50 µH (0.9 %)	1592 Hz	1669 Hz	0.42	0.18	1.17 A (0.51 %)	0.052
75 µH (1.4 %)	1299 Hz	1394 Hz	0.35	0.12	0.74 A (0.32 %)	0.044
100 µH (1.8 %)	1125 Hz	1233 Hz	0.31	0.08	0.54 A (0.23 %)	0.039

At the nominal 50 µH the line is 0.51 % of I_n: inside the 1.0 % of its own grid, at the edge of the 0.5 % used in Example 1. The attenuation is however entirely the grid's: a stiffer PCC (25 µH, e.g. a larger transformer) doubles the harmonic to 1.23 % and fails both limits, a weaker one improves it. The same filter at $f_{sw} = 2\,\text{kHz}$ would have its lines at 1900 and 2100 Hz on the resonance, $k_a = 2.3$ and 1.3 (7 % and 4 % of I_n): the reason why Example 1 needs L_2.

D4 – Resonance and damping. $f_{res} = 1669\,\text{Hz} = 0.42 f_{sw}$ lies above $1.2 f_s/6 = 1600\,\text{Hz}$, but only just: for $L_g > 49\,\mu\text{H}$ the resonance enters the band 1067–1600 Hz around $f_s/6$ and at 100 µH it is at 1233 Hz, below $f_s/6$, where i_1 feedback is unstable without damping [6]. The 50 mΩ give $\zeta = 0.04$–0.07 ($Q = 7$–12) with $P_{R_d} = 3R_d(I_c^2 + I_{rip,rms}^2) = 118\,\text{W}$ and an attenuation penalty $\sqrt{1+(\omega_{sw} R_d C)^2} = 1.03$: a correct backstop at the nominal grid, the only stabiliser at a weak one, to be verified with the controller gains and the 188 µs delay.

D5 – Aligned version. With the criteria of Example 1 (capacitor within the 5 % bound, L_2 as a design variable, resonance band inside $[1.2 f_s/6,\ f_{sw}/2]$ for $L_g = 25$–$100\,\mu\text{H}$, R–C branch with $n = 0.25$), the same $L_1 = 500\,\mu\text{H}$ gives

$$C_1 = 60\,\mu\text{F}\ (3.3\,\%), \qquad L_2 = 140\,\mu\text{H}\ (2.5\,\%),$$

$$f_{res} = \frac{1}{2\pi}\sqrt{\frac{L_1 + L_2'}{L_1 L_2' C_1}} = 1845\,\text{Hz}\ (L_g = 25\,\mu\text{H}), \qquad 1751\,\text{Hz}\ (50\,\mu\text{H}), \qquad 1614\,\text{Hz}\ (100\,\mu\text{H}),$$

$$k_a = \frac{1}{\omega^2 L_2' C_1 - 1} = 0.18,\ 0.15,\ 0.12 \ \Rightarrow\ I_2(f_{sw}+2f_1) = 1.15,\ 0.97,\ 0.75\,\text{A}\ (0.50\text{–}0.32\,\%\ I_n),$$

$$L_p = L_1 \parallel L_2' = 124\,\mu\text{H}, \qquad R_0 = \sqrt{L_p/C_1} = 1.44\,\Omega, \qquad C_2 = 0.25\,C_1 = 15\,\mu\text{F},$$

$$R_2 = \rho_P R_0 = 4.5 \cdot 1.44 = 6.5\,\Omega, \qquad \zeta = 0.06, \qquad P_{R_2} = 68 + 20 = 88\,\text{W},$$

with the total capacitance at 4.1 % of C_b ($I_c = 7.5\,\text{A}$ instead of 25 A), the drop of $L_1 + L_2 = 11.6\,\%$ in quadrature ($V_1 = 401\,\text{V}$, +0.7 %, $m = 1.03$) and the attenuation of the original design at its nominal grid now guaranteed over the whole L_g range, with the resonance pinned between 0.40 and $0.46 f_{sw}$. Keeping $C = 200\,\mu\text{F}$ and adding L_2 is not an option: $f_{res} \geq 1.2 f_s/6$ at L_g^{max} would require $L_1 \parallel L_2' \leq 49\,\mu\text{H}$, i.e. no room for any L_2; the large capacitor and the absence of L_2 go together.

Review of the 690 V design in one paragraph. What is already aligned: L_1 at the 25 % ripple allowance, the 50 mΩ at $\zeta \approx 0.05$ with negligible loss and attenuation penalty, and the line at $f_{sw} + 2f_1$ within the limit of its grid at 4 kHz. What the criteria of this note change: the capacitor from 11 % to 3.3 % (plus 0.8 % in the damping branch), a 2.5 % line reactor ($L_2 = 140\,\mu\text{H}$) that makes attenuation and resonance independent of the PCC, and the R–C branch (15 µF, 6.5 Ω) in place of the series resistor. The price is the reactor and the 11.6 % series reactance (in quadrature, hence harmless for the voltage); the gain is the reactive

power (30 kvar to 11 kvar), a resonance that cannot drift onto $f_s/6$ or onto the switching cluster when the transformer changes, and a design that would survive a move to 2 kHz with the method of Example 1 rather than fail silently.

6.4 Example 2: L(RC) sine filter between inverter and motor

Inputs. $P_n = 450\,\text{kW}$, $V = 480\,\text{V}$, $f_{1,max} = 60\,\text{Hz}$ (rated speed, no field weakening), $V_{dc} = 800\,\text{V}$, $f_{sw} = 2\,\text{kHz}$, SVM, motor cable $l = 150\,\text{m}$ with $v = 150\,\text{m/µs}$, inverter edge $t_r = 0.3\,\text{µs}$ (1200 V IGBT modules). Two machines of the same rating:

- IM, 4-pole, 1780 rpm: $I_n = 650\,\text{A}$ ($\cos\varphi = 0.87$, $\eta = 0.96$), $\sigma L_s = 0.18\,\text{pu}$, $R_{hf} = 0.10\,\text{pu}$ (deep-bar rotor at 700 Hz);
- PMSM (IPM), 8-pole, 900 rpm: $I_n = 590\,\text{A}$ ($\cos\varphi = 0.95$, $\eta = 0.97$), $L_q = 0.30\,\text{pu}$ ($L_d \approx 0.20\,\text{pu}$), $R_{hf} = 0.02\,\text{pu}$.

Base quantities with $Z_b = V_{ph}/I_n$: IM $Z_b = 0.426\,\Omega$, $L_b = 1.131\,\text{mH}$, $C_b = 6.22\,\text{mF}$, $L_{hf} = 0.18 L_b = 204\,\text{µH}$; PMSM $Z_b = 0.470\,\Omega$, $L_b = 1.246\,\text{mH}$, $C_b = 5.65\,\text{mF}$, $L_{hf} = L_q = 374\,\text{µH}$.

B1 – Is a filter needed, and which. From (5.1):

$$l_{crit} = \frac{v\,t_r}{2} = \frac{150 \cdot 0.3}{2} = 22.5\,\text{m} \ll l, \qquad \hat{V}_m = V_{dc}(1+\Gamma_m) = 800 \cdot 1.8 = 1440\,\text{V} \quad (\Gamma_m = 0.8),$$

within the NEMA MG 1-31 envelope of an inverter-duty motor ($3.1V = 1488\,\text{V}$) but far above the 1000 V of a general-purpose motor (MG 1-30). A dv/dt filter limiting the overshoot to $OF = 0.2$ would need $l_{crit} \geq l\Gamma_m/OF = 600\,\text{m}$, i.e. $t_r \geq 8\,\text{µs}$ ($f_0 \leq 38\,\text{kHz}$, $dv/dt \leq 100\,\text{V/µs}$) with $C_f \leq 0.22\,\text{µF}$ (loss bound $3 f_{sw} C_f V_{dc}^2 = 845\,\text{W}$). The sine filter is chosen: standard motor, long cable, and at 2 kHz the sinusoidal voltage also removes the audible switching noise and the harmonic losses of the machine.

B2 – Resonance window. From (5.2):

$$8 f_{1,max} = 480\,\text{Hz} \leq f_0 \leq \frac{f_{sw}}{4} = 500\,\text{Hz}\ \left(\frac{f_{sw}}{3} = 667\,\text{Hz}\right), \qquad \frac{f_{sw}}{f_{1,max}} = 33.3 \text{ (the window needs 24–40).}$$

The window is at its limit: the design places the *loaded* resonance at the relaxed bound $f_p = f_{sw}/3 = 667\,\text{Hz}$ and derives $f_0 = f_p/\sqrt{1 + L_f/L_{hf}}$ from it. Any field weakening (e.g. 72 Hz) would require $f_0 \geq 576\,\text{Hz}$ and close the window; the dead-time harmonics are checked in B5.

B3 – Inductor L_f. The drop bound of (5.3) and the ripple bound conflict at 2 kHz:

$$\text{drop } \epsilon_L \leq 0.08: \quad L_f \leq \frac{0.08 \cdot 277}{377 \cdot 650} = 90\,\text{µH} \;\Rightarrow\; \Delta i_{\text{pp},max} = \frac{V_{dc}}{6 f_{sw} L_f} = 741\,\text{A} = 80\,\%\ \hat{I}_n \text{ (not acceptable);}$$

$$\text{ripple } k_r = 0.5: \quad L_f \geq \frac{V_{dc}}{6 f_{sw} k_r \hat{I}_n} = \frac{800}{6 \cdot 2000 \cdot 0.5 \cdot 919} = 145\,\text{µH} \;\Rightarrow\; L_f = 150\,\text{µH}\ (13.3\,\%),$$

the ripple bound wins and the drop (13.3 % at 60 Hz) is budgeted in V_{dc} (step B4). Then

$$\Delta i_{\text{pp},max} = \frac{800}{6 \cdot 2000 \cdot 150 \cdot 10^{-6}} = 444\,\text{A}\ (48\,\%), \qquad I_{rip,rms} = \frac{800}{39.8 \cdot 2000 \cdot 150 \cdot 10^{-6}} = 67\,\text{A}\ (10.3\,\%),$$

$$I_i(f_{sw}\text{ cluster}) = \frac{800}{42.2 \cdot 2000 \cdot 150 \cdot 10^{-6}} = 63\,\text{A},$$

with $k = 1/6$ and $1/39.8$ at full modulation (Table 2, 5). PMSM: $L_f \geq 160\,\text{µH} \Rightarrow L_f = 160\,\text{µH}$ (12.8 %), $\Delta i_{\text{pp},max} = 417\,\text{A}$ (50 %), $I_{rip,rms} = 63\,\text{A}$ (10.7 %).

B4 – Capacitor C_f from the loaded resonance. With $f_p = 667$ Hz and (3.4), (5.4):

$$\text{IM:}\qquad f_0 = \frac{f_p}{\sqrt{1 + L_f/L_{hf}}} = \frac{667}{\sqrt{1 + 150/204}} = 506\,\text{Hz}\ (8.4 f_{1,max}),$$

$$C_f = \frac{1}{(2\pi f_0)^2 L_f} = \frac{1}{(2\pi \cdot 506)^2 \cdot 150 \cdot 10^{-6}} = 660\,\mu\text{F}\ (10.6\,\%\ C_b), \qquad I_c = \omega_1 C_f V_{ph} = 69\,\text{A}\ (10.6\,\%);$$

$$\text{PMSM:}\qquad f_0 = \frac{667}{\sqrt{1 + 160/374}} = 558\,\text{Hz}\ \Rightarrow\ C_f = 509\,\mu\text{F} \to 500\,\mu\text{F}\ (8.9\,\%),$$

$$f_0 = 563\,\text{Hz}\ (9.4 f_{1,max}), \qquad f_p = 672\,\text{Hz}, \qquad I_c = 52\,\text{A}\ (8.9\,\%).$$

The larger L_{hf} of the PMSM shifts the resonance less ($f_p/f_0 = 1.19$ against 1.32) and allows the smaller capacitor. Inverter operating point at rated load from the phasor relations of (5.4) (stator current lagging, capacitor current leading): IM $I_i = 619$ A (power factor 0.91 at the inverter), $V_i = 293$ V phase (508 V line), $m = 1.04$; PMSM $I_i = 576$ A, $V_i = 287$ V, $m = 1.02$: both inside the SVM range of $V_{dc} = 800$ V and below the inverter rating, the capacitor current partly compensating the lagging stator current.

B5 – Attenuation, machine damping, dead-time harmonics. From (3.4):

$$\left|\frac{v_c}{v_i}\right|_{f_{sw}} = \frac{1}{(f_{sw}/f_p)^2 - 1} = \frac{1}{9 - 1} = 0.125\ (-18.1\,\text{dB}),$$

$$v_{c,sw} \approx \frac{I_i(f_{sw})}{\omega_{sw} C_f} = \frac{63}{2\pi \cdot 2000 \cdot 660 \cdot 10^{-6}} = 7.6\,\text{V}\ (2.8\,\%\ V_{ph}),$$

PMSM 0.127 (−17.9 dB), 9.4 V (3.4 %): a weak sine filter by the standards of (5.2), but enough for the insulation and the machine losses. Machine damping with $L_p = L_f \parallel L_{hf}$: IM $L_p = 86\,\mu$H, $G_m = R_{hf}/(R_{hf}^2 + (2\pi f_p L_{hf})^2) = 58$ mS, $\zeta_m = \frac{1}{2} G_m \sqrt{L_p/C_f} = 0.011$; PMSM $L_p = 112\,\mu$H, $G_m = 3.8$ mS, $\zeta_m = 0.0009$. The 5th and 7th dead-time harmonics at 300 and 420 Hz are amplified by $1/|1 - (f/f_p)^2| = 1.25$ and 1.66: acceptable with dead-time compensation and the damping of B6.

B6 – Damping, R–C branch added to C_f. With $R_0 = \sqrt{L_p/C_f}$ (IM 0.362 Ω, PMSM 0.473 Ω) and the fundamental at $f_{1,max}$ in (4.8):

	IM						PMSM					
n	C_2	R_2	ζ	P_{R_2}	P_{R_2}/P_n	I_c total	C_2	R_2	ζ	P_{R_2}	P_{R_2}/P_n	I_c total
0.25	165 µF	1.63 Ω	0.059	1.53 kW	0.34 %	86 A (13 %)	125 µF	2.13 Ω	0.059	1.19 kW	0.26 %	65 A (11 %)
0.5	330 µF	0.90 Ω	0.112	3.35 kW	0.74 %	103 A (16 %)	250 µF	1.18 Ω	0.112	2.59 kW	0.57 %	78 A (13 %)
1	660 µF	0.54 Ω	0.204	7.87 kW	1.75 %	138 A (21 %)	500 µF	0.71 Ω	0.204	6.02 kW	1.34 %	104 A (18 %)
R_d	$1/(3 \cdot 2\pi f_p C_f) = 121$ mΩ, $\zeta = 0.17$: 3.35 kW (0.74 %), attenuation ×1.41						158 mΩ, $\zeta = 0.17$: 3.16 kW (0.70 %), attenuation ×1.41					

The loss is almost entirely the fundamental term (the switching term is 90–310 W), because the capacitor is 9–11 % of C_b and the branch carries $\omega_1 C_2 V_{ph}$ at full voltage all the time; it scales with n. Passive damping to $\zeta = 0.2$ costs 1.3–1.8 % of P_n and is not acceptable; the series resistor is no better and also degrades the attenuation. Selected: $n = 0.25$ ($\zeta = 0.06$, $Q \approx 8$, 0.26–0.34 %) as a backstop for the V/f mode and the dead-time harmonics, with the bulk of the damping provided actively by the cascaded control of Section 5.6.

B7 – Ratings and control. IM: L_f 650 A RMS plus 67 A RMS ripple at 2 kHz, saturation $\geq \hat{I}_n + \Delta i_{\text{pp},max}/2 = 1141$ A; $C_f \geq 528$ V AC plus 11 V peak ripple, $\sqrt{I_c^2 + I_{rip,rms}^2} = 96$ A RMS; C_2 17 A, R_2 1.5 kW. PMSM: saturation ≥ 1043 A, capacitor 82 A, C_2 13 A, R_2 1.2 kW. Control (both): inverter-current loop $\alpha_i \leq 2\pi f_{sw}/10 = 2\pi \cdot 200$ Hz, capacitor-voltage loop $\alpha_i/4$, stator-current loop $\leq f_p/5 = 133$ Hz, HF injection $f_{inj} \leq f_p/3 = 222$ Hz (in practice disabled); for the IPM the observer uses $L_d \neq L_q$, so f_p is 672 Hz on the q axis and ≈ 700 Hz on the d axis.



6.5 Comparison: AFE, induction motor, PMSM

Table 6: The three L(RC) sine filters of the 450 kW, 480 V, 2 kHz drive (SVM, $V_{dc} = 800$ V). Percentages are per unit of the respective converter base; losses refer to S_n (AFE) or P_n (motors).

	AFE, grid side (L(RC)L)	IM (L(RC))	PMSM, IPM (L(RC))
rating, current	500 kVA, 601 A	450 kW, 650 A	450 kW, 590 A
design driver	IEEE 519 line limit at $h = 35$ (attenuation)	resonance window at pulse ratio 33 and ripple	same, with the larger L_{hf}
converter-side inductance	$L_1 = 180$ µH (14.7 %)	$L_f = 150$ µH (13.3 %)	$L_f = 160$ µH (12.8 %)
second inductance	$L_2 = 180$ µH (14.7 %) $+L_g =$ 13–51 µH	$L_{hf} = \sigma L_s = 204$ µH (18 %), given	$L_{hf} = L_q = 374$ µH (30 %), given, $L_d \neq L_q$
main capacitor C_1	280 µF (4.9 %)	660 µF (10.6 %)	500 µF (8.9 %)
R–C branch ($n = 0.25$)	70 µF, 2.59 Ω, $\zeta = 0.06$	165 µF, 1.63 Ω, $\zeta = 0.06$	125 µF, 2.13 Ω, $\zeta = 0.06$
damping loss	550 W (0.11 %)	1.53 kW (0.34 %)	1.19 kW (0.26 %)
damping by the load	none (L_g lossless)	$\zeta_m = 0.011$ (rotor cage)	$\zeta_m = 0.001$
resonance	f_{res} = 946–986 Hz (0.47–0.49f_{sw}), set by L_g	$f_0 = 506$ Hz, $f_p = 667$ Hz ($f_{sw}/3$), shift +32 %	$f_0 = 563$ Hz, $f_p = 672$ Hz ($f_{sw}/3$), shift +19 %
attenuation at f_{sw}	$k_a = 0.11$ (−19 dB) on i_2/i_1	0.125 (−18 dB) on v_c/v_i	0.127 (−18 dB)
residual at the far side	$I_2(f_{sw}+2f_1) = 2.7$ A (0.45 %)	$v_{c,sw} = 7.6$ V (2.8 %)	9.4 V (3.4 %)
ripple, converter side	328 A pp (39 %), 50 A RMS	444 A pp (48 %), 67 A RMS	417 A pp (50 %), 63 A RMS
fundamental drop	29.5 % (quadrature, +4 % voltage)	13.3 % ($m = 1.04$)	12.8 % ($m = 1.02$)
capacitor current (C_1)	29 A (4.9 %)	69 A (10.6 %)	52 A (8.9 %)
inductor peak current	1015 A	1141 A	1043 A
stability constraint	$f_{res} > f_s/6 = 667$ Hz with i_1 feedback	f_p inside the current loop: cascaded control or observer	same, anisotropic plant
current-loop bandwidth	$f_c \leq f_{res}/3 = 315$ Hz	stator loop $\leq f_p/5 = 133$ Hz	134 Hz
sensorless / PLL	PLL ≤ 20 Hz at $S_{sc,min}$	flux observer with the filter model	back-EMF observer with the filter model; HF injection $\leq$ 220 Hz
what it protects	PCC harmonics, grid resonance	motor insulation (1440 V peak without filter), losses, noise	same, plus magnet losses and demagnetisation margin
how to lighten it	trap $L_t = 22.6$ µH, or 4 kHz (11 % total L; at 4 kHz also the grid as L_2, Section 6.3)	4 kHz: window opens, L_f and C_f halve	same; lower pole count would need higher f_{sw}

Table 6 shows what is common and what differs. The three filters share the inductor (13–15 %, set by a ripple of 40–50 % at 2 kHz), the resonance placement at a fixed fraction of f_{sw} ($f_{sw}/2$ for the AFE with i_1 feedback, $f_{sw}/3$ for the motors), the modest attenuation of 18–19 dB that follows from it, and the conclusion that at pulse ratio 33 the passive damping must be kept small ($\zeta \approx 0.06$) and complemented by active damping. They differ in what fixes the second inductance: on the grid side L_2 is a design variable and L_g an uncertainty, so the resonance is a band and the harmonic limit sets the heaviest element of the whole drive (30 % of inductance, or a trap); on the motor side the second inductance is the machine itself, so the capacitor is the design variable, the resonance shift is known from L_{hf}, and the capacitor is 9–11 % with the fundamental current and the damping loss that go with it. Between the two machines, the PMSM has the larger L_{hf} (30 % against 18 %) and therefore the smaller capacitor, the smaller resonance shift and the lower damping loss, but it gives no damping of its own, its plant is anisotropic ($L_d \neq L_q$, saturating with i_q) and its sensorless control must carry the filter model; the IM is more forgiving on the control side and pays with the larger capacitor. For both motors the stator-current bandwidth of about 130 Hz is the real cost of the sine filter at 2 kHz: adequate for pumps, fans, compressors and propulsion, not for a servo axis.

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